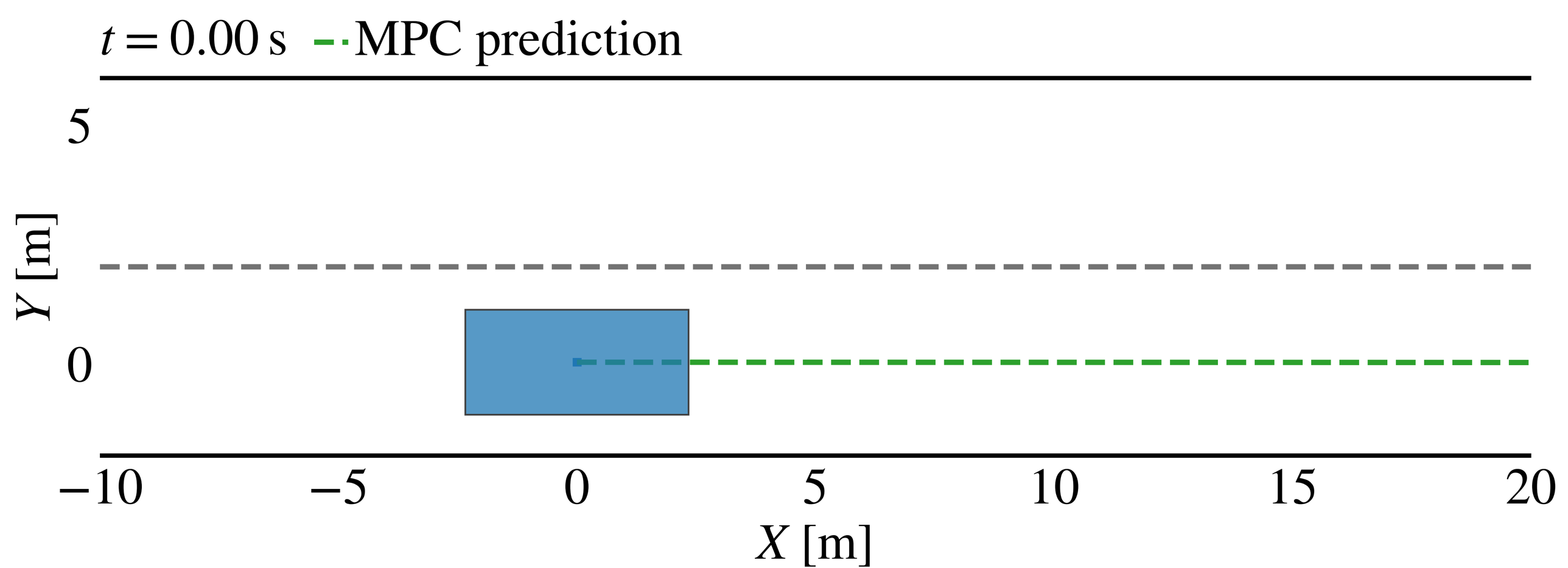
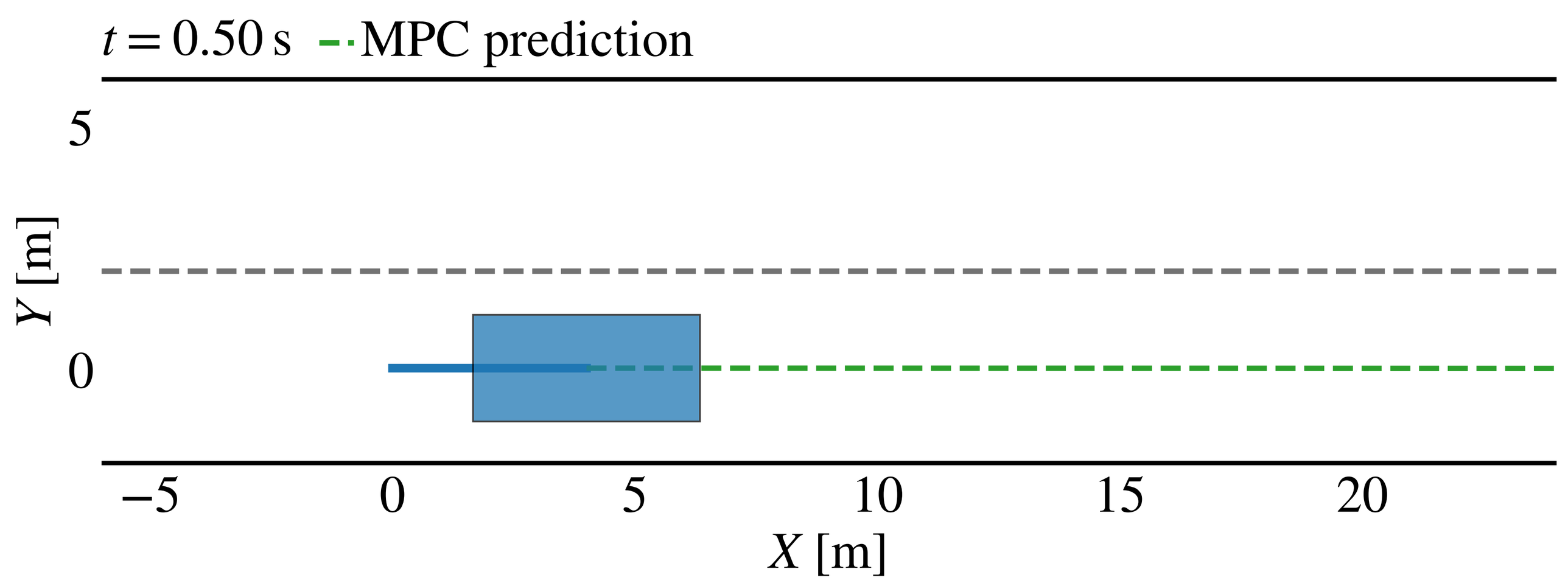
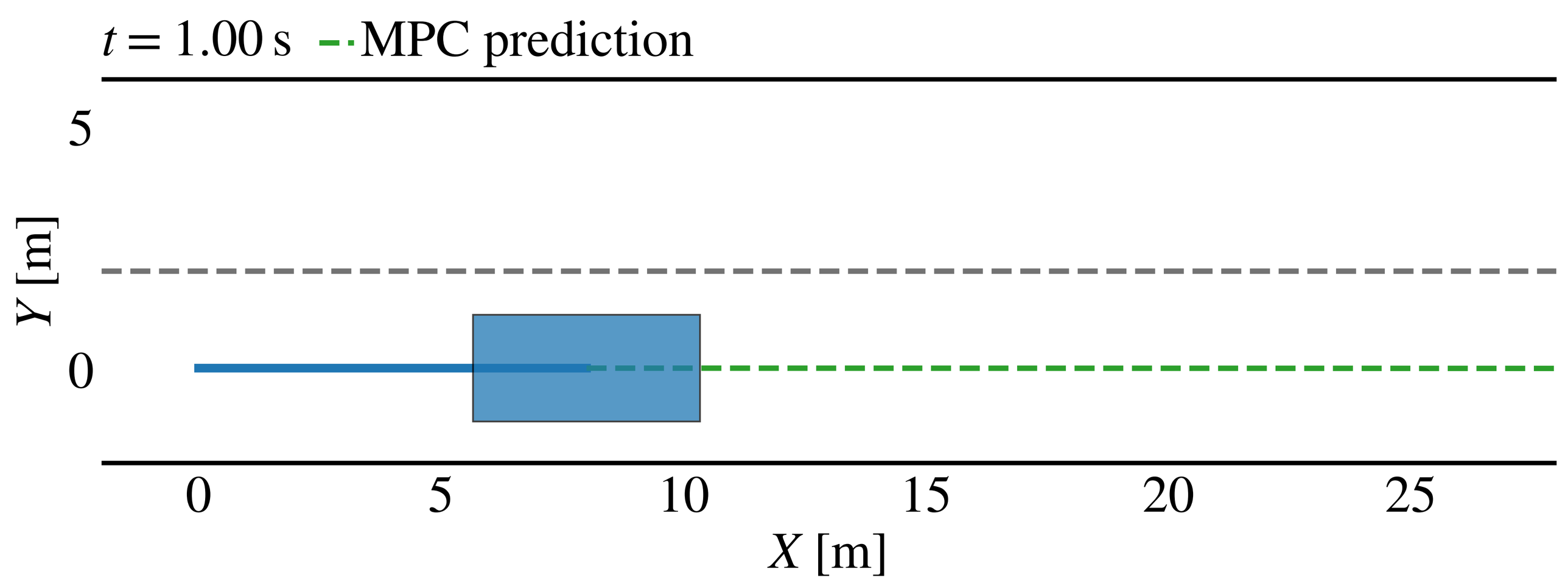
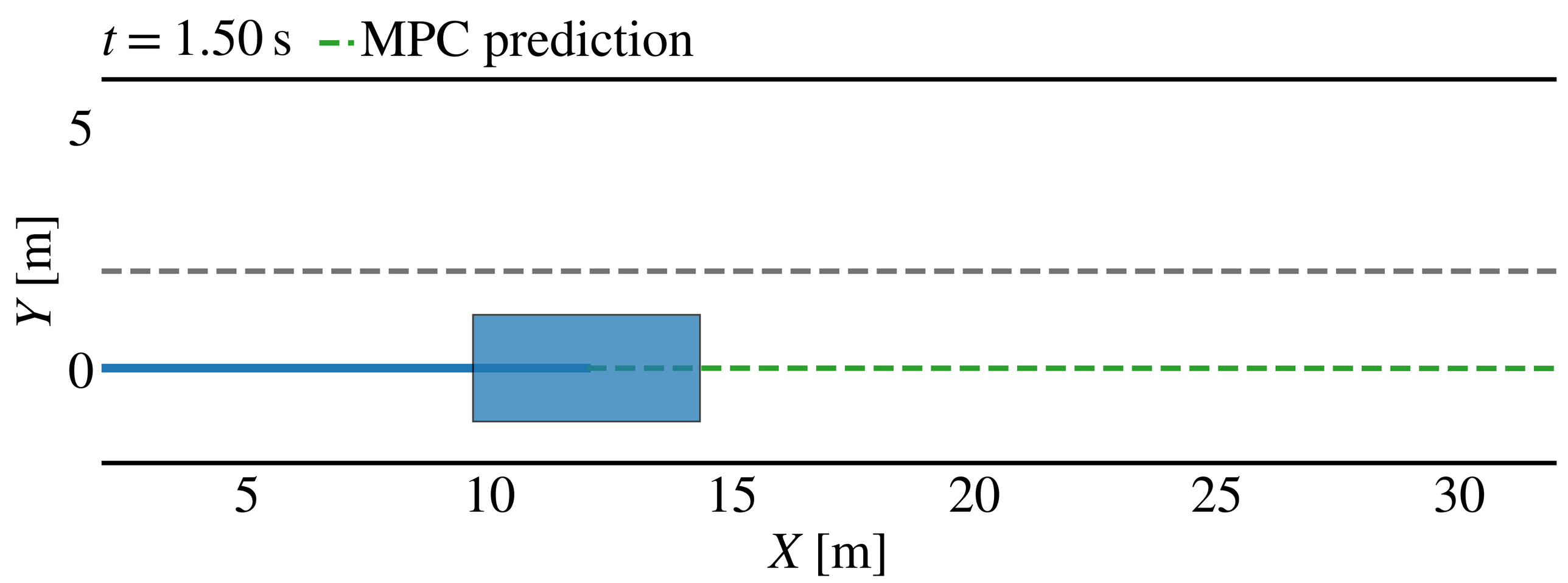
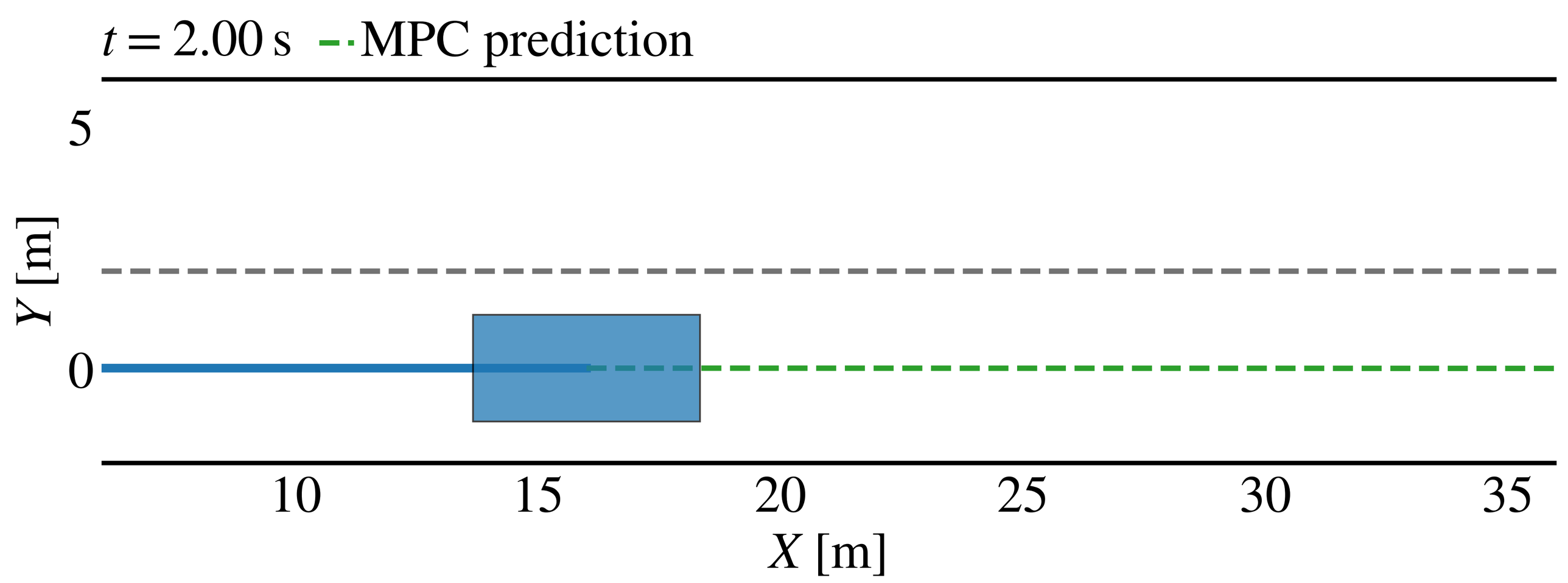
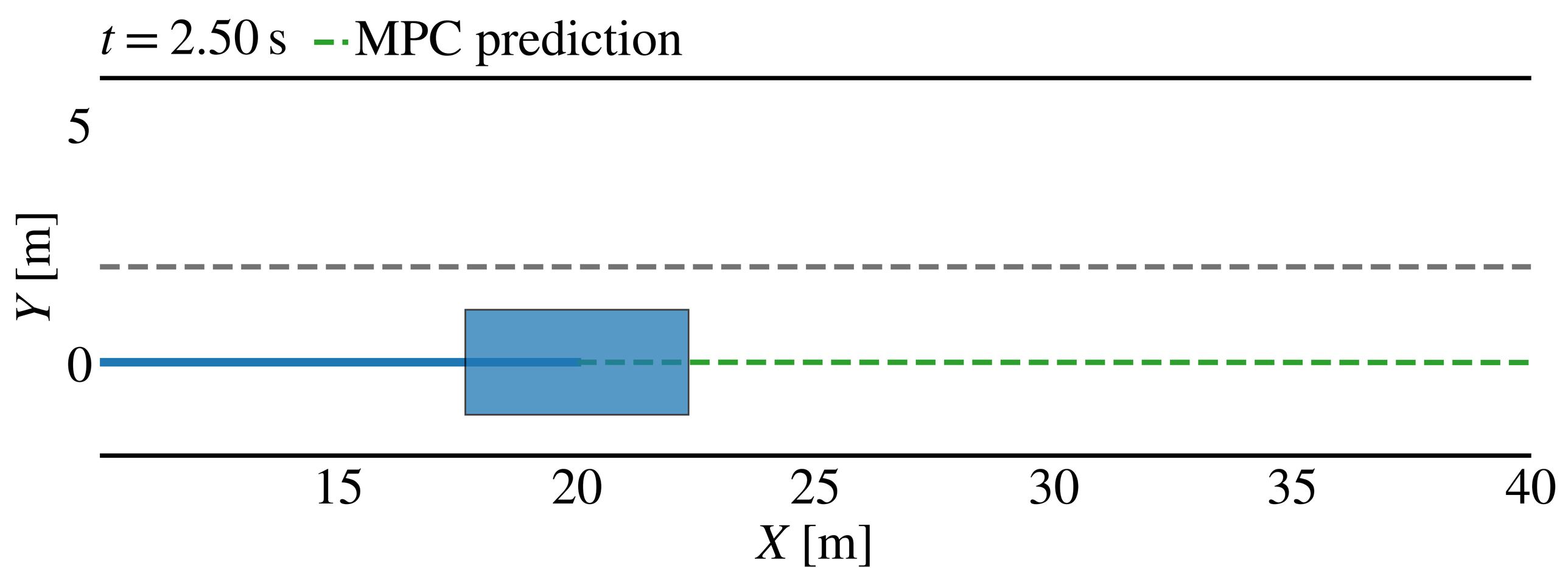

$t = 0.00$ s
MPC prediction
5
0
Y [m]
−10
−5
0
5
10
15
20
X [m]

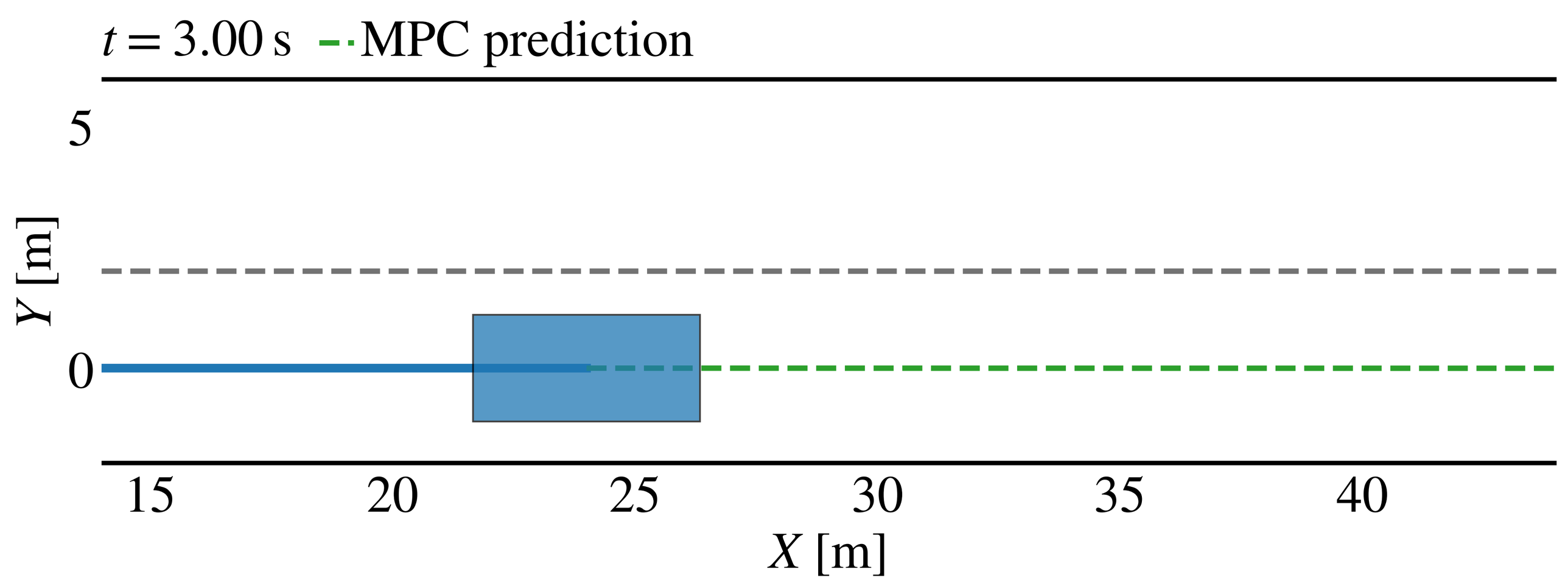

$t = 0.50$ s
MPC prediction
5
0
Y [m]
−5
0
5
10
15
20
X [m]

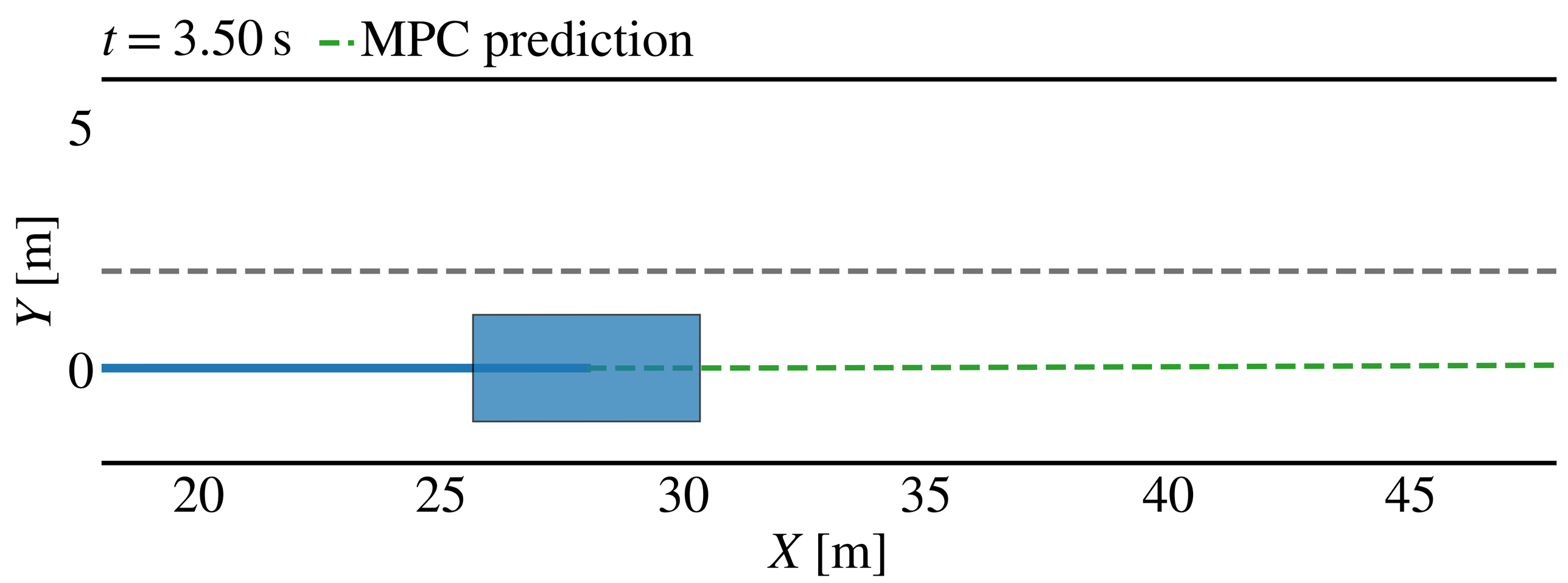

$t = 1.00$ s
MPC prediction
5
0
Y [m]
0
5
10
15
20
25
X [m]

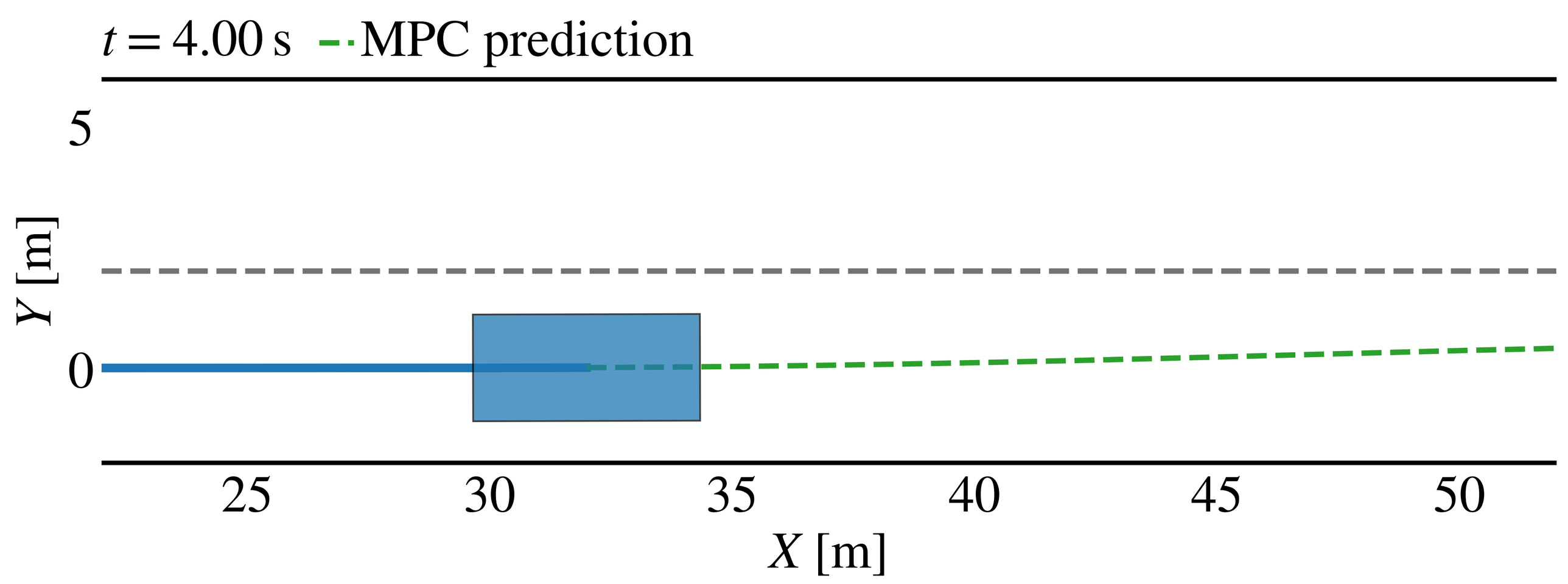

$t = 1.50$ s
MPC prediction
5
0
Y [m]
5
10
15
20
25
30
X [m]

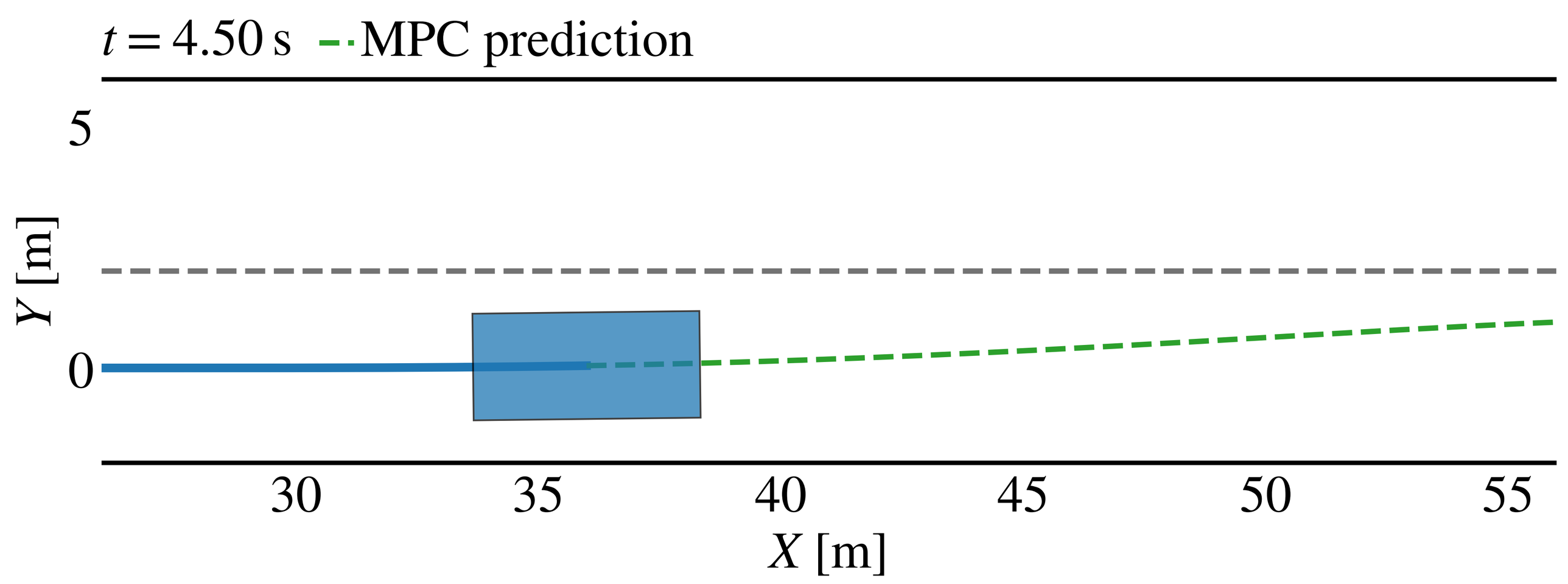

$t = 2.00$ s
MPC prediction
5
0
Y [m]
10
15
20
25
30
35
X [m]

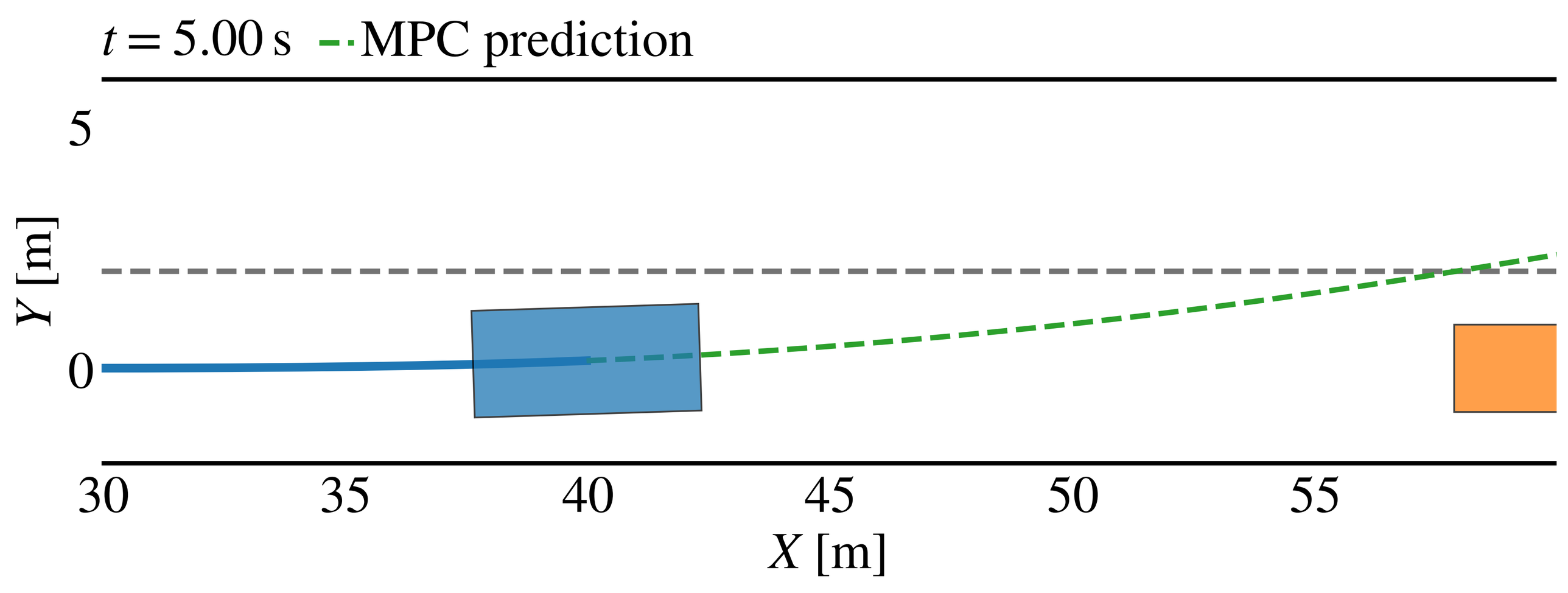

$t = 2.50$ s
MPC prediction
5
0
Y [m]
15
20
25
30
35
40
X [m]

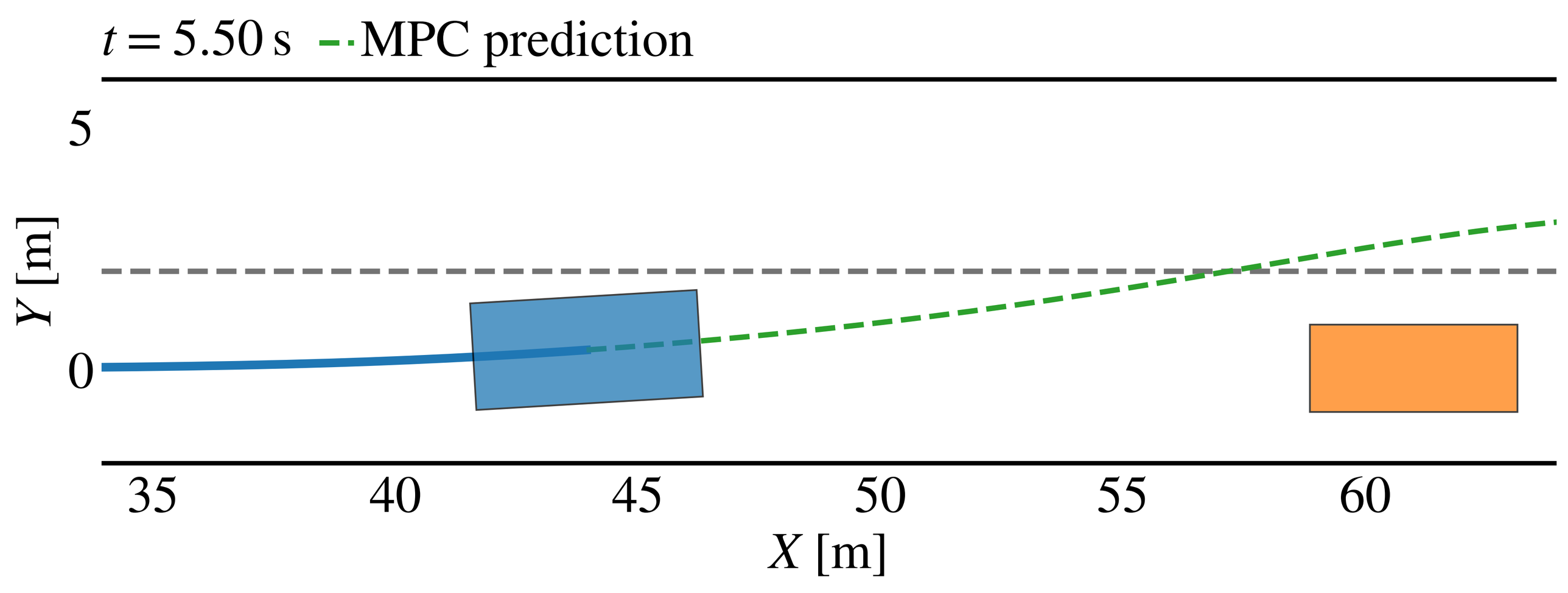

$t = 3.00$ s
MPC prediction
5
0
Y [m]
15
20
25
30
35
40
X [m]

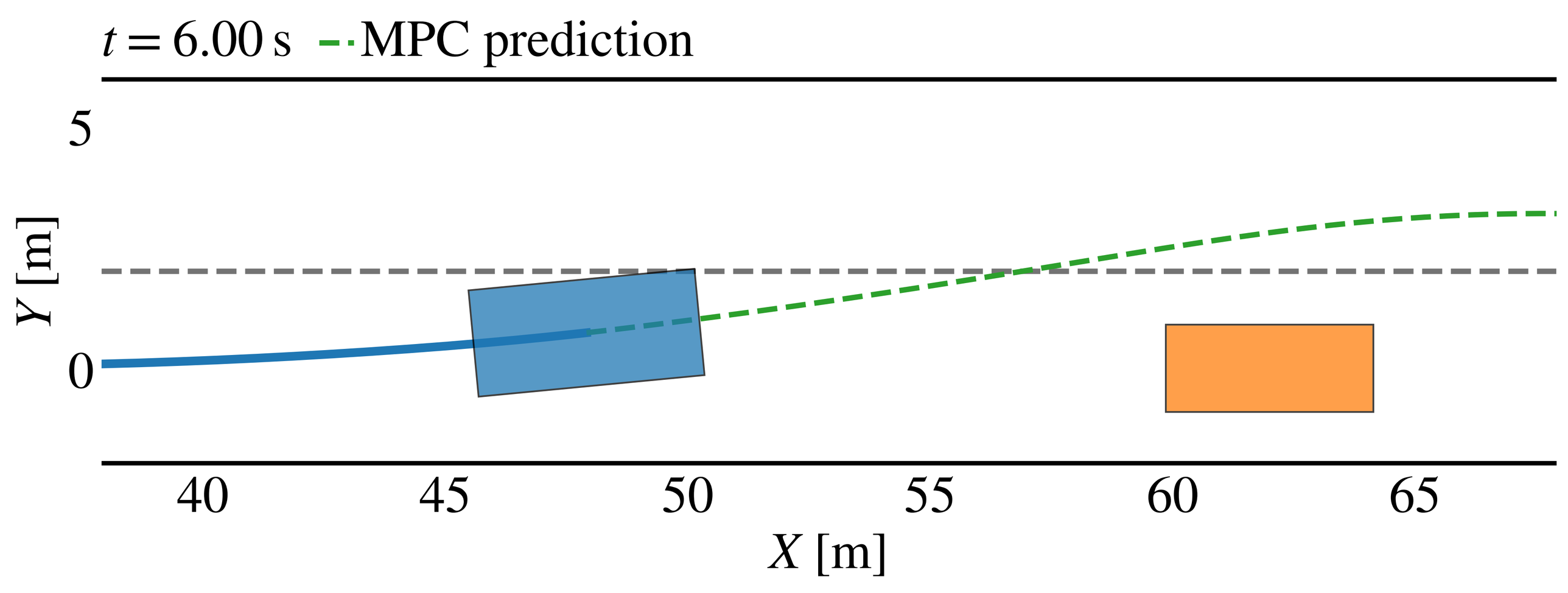

$t = 3.50$ s
MPC prediction
5
0
Y [m]
20
25
30
35
40
45
X [m]

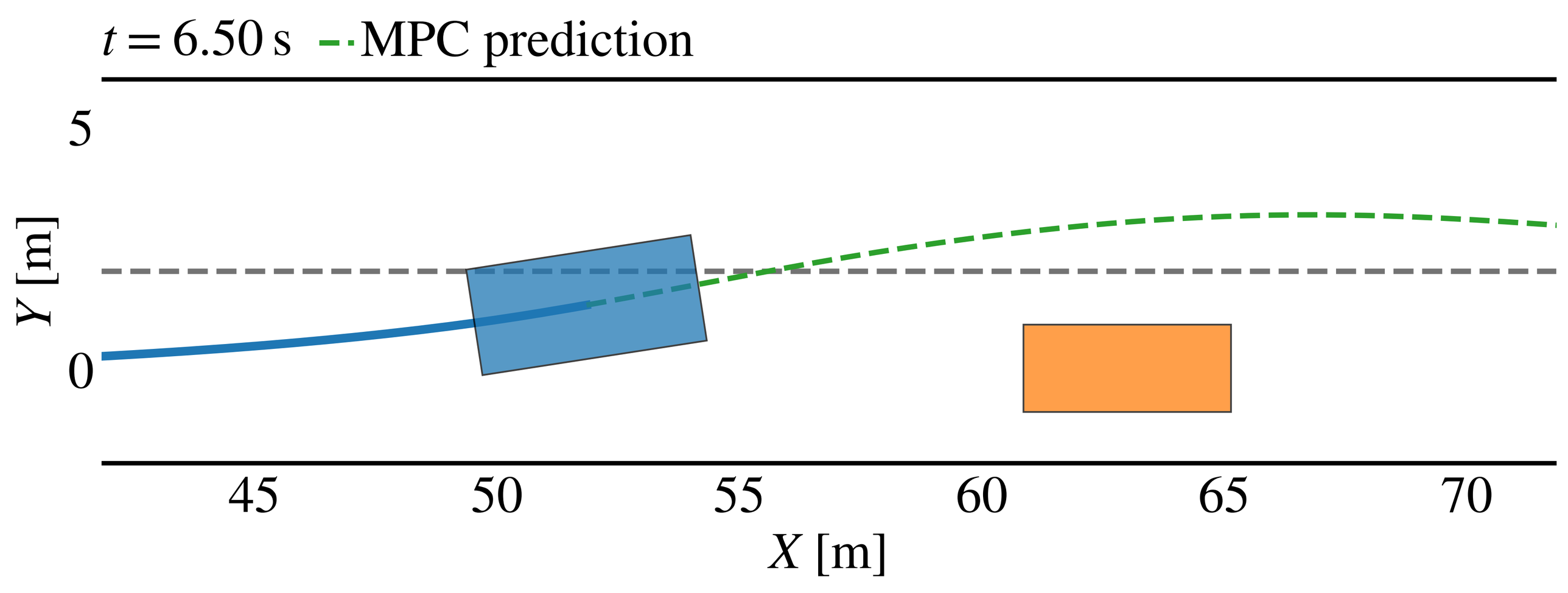

$t = 4.00$ s
MPC prediction
5
0
Y [m]
25
30
35
40
45
50
X [m]

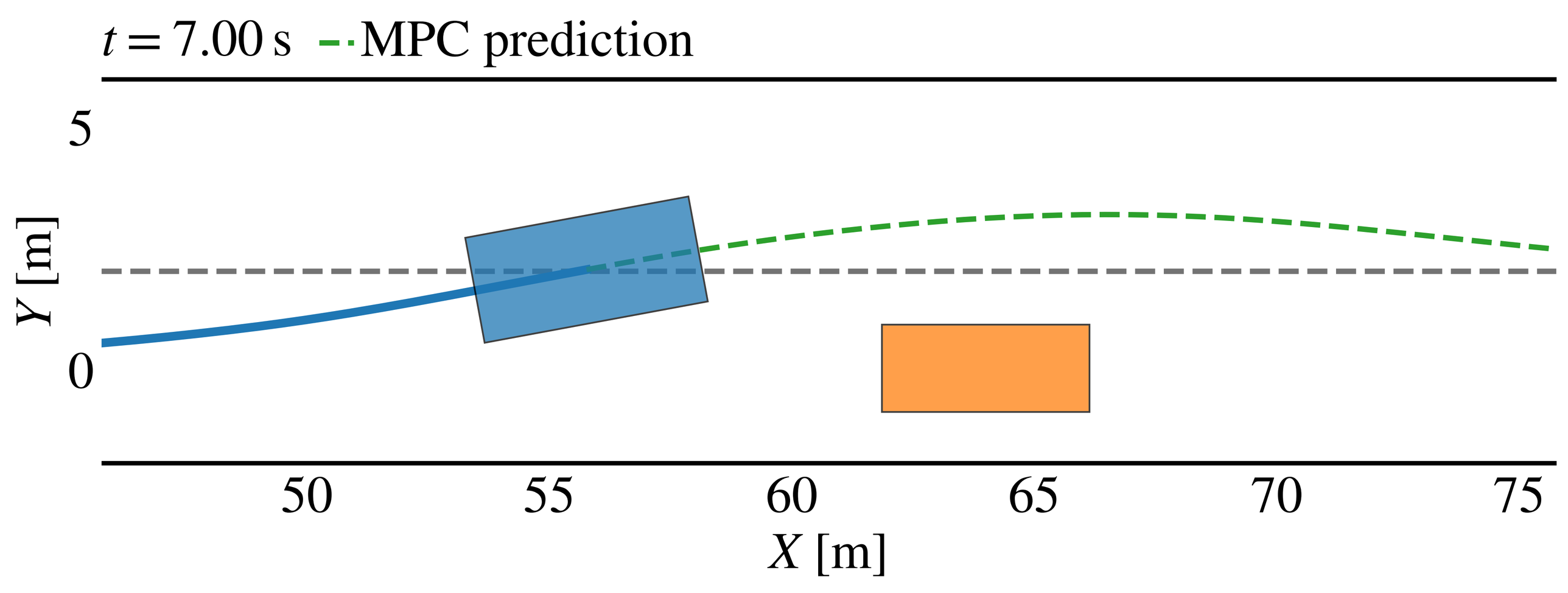

$t = 4.50$ s
MPC prediction
5
0
Y [m]
30
35
40
45
50
55
X [m]

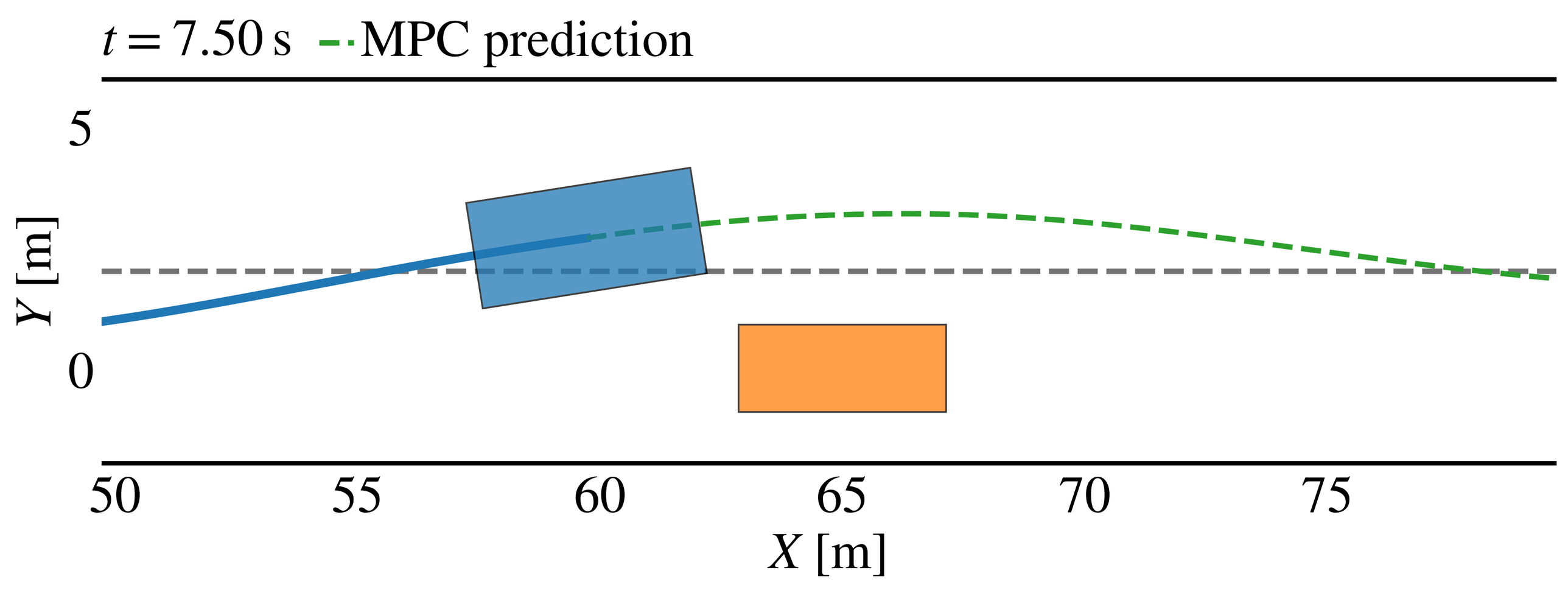

$t = 5.00$ s
MPC prediction
5
0
Y [m]
30
35
40
45
50
55
X [m]

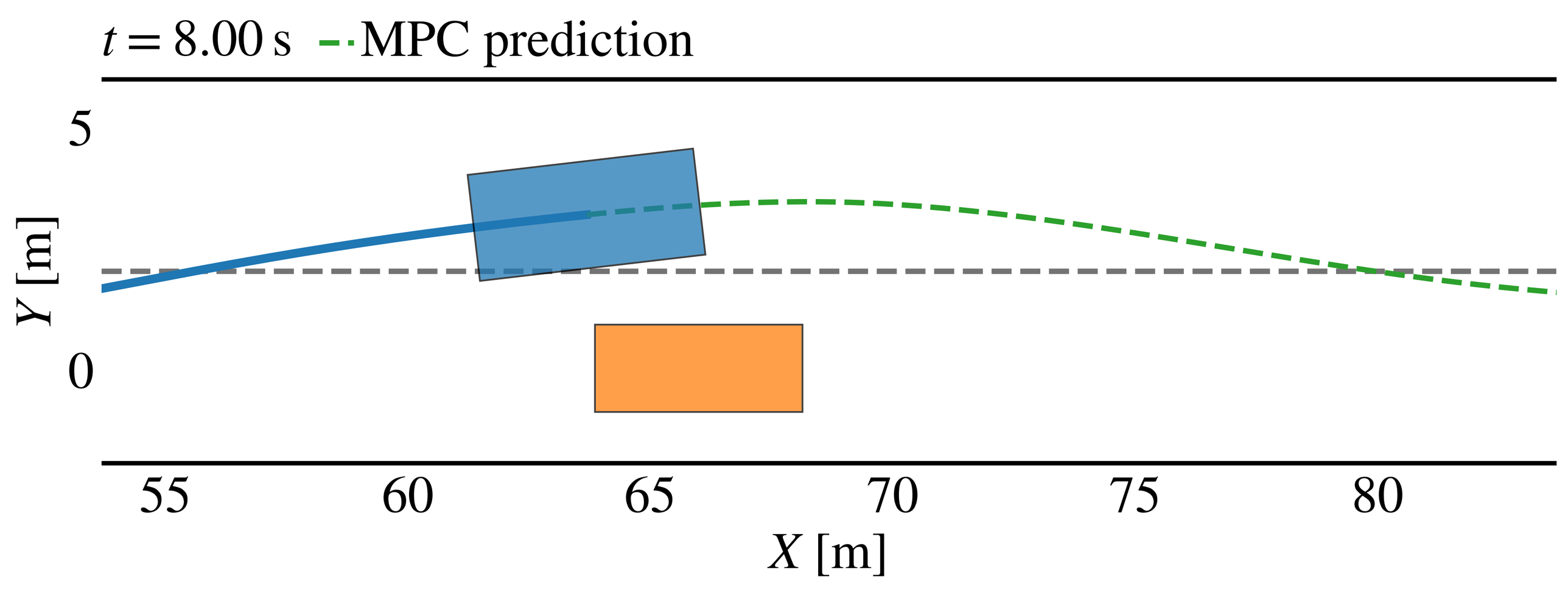

$t = 5.50$ s
MPC prediction
5
0
Y [m]
35
40
45
50
55
60
X [m]

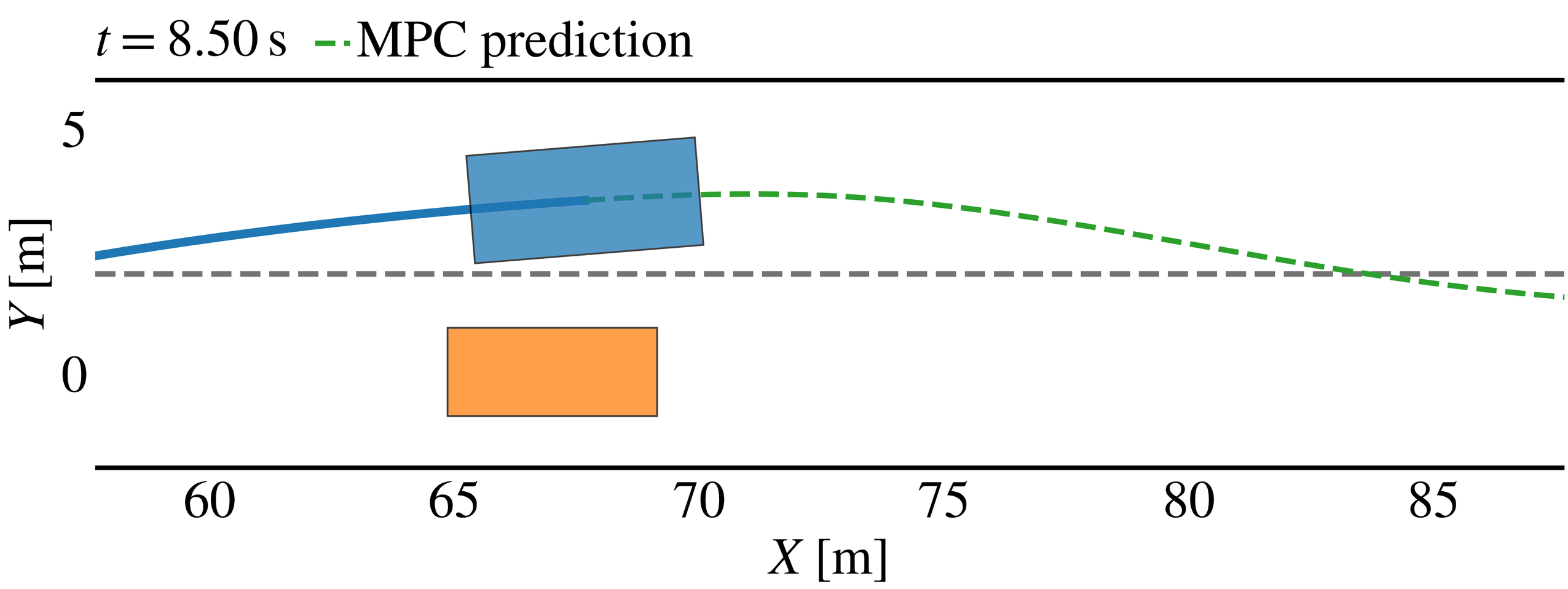

$t = 6.00$ s
MPC prediction
5
0
Y [m]
40
45
50
55
60
65
X [m]

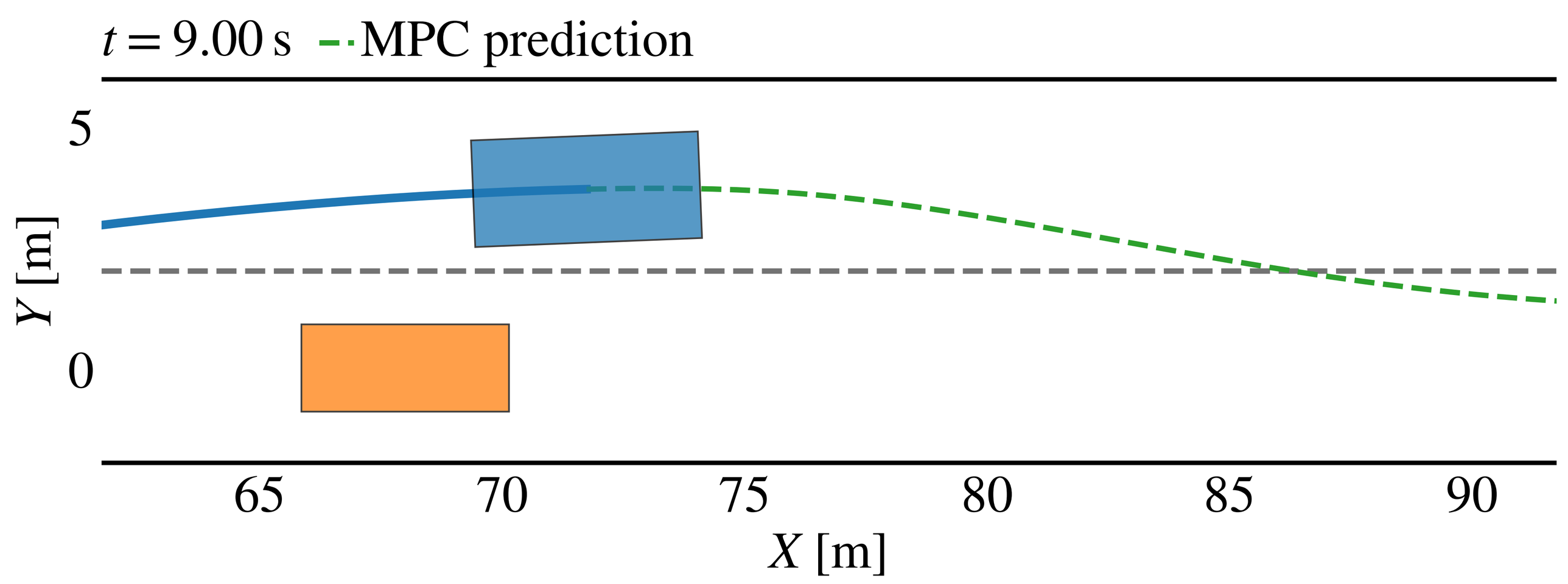

$t = 6.50$ s
MPC prediction
5
0
Y [m]
45
50
55
60
65
70
X [m]

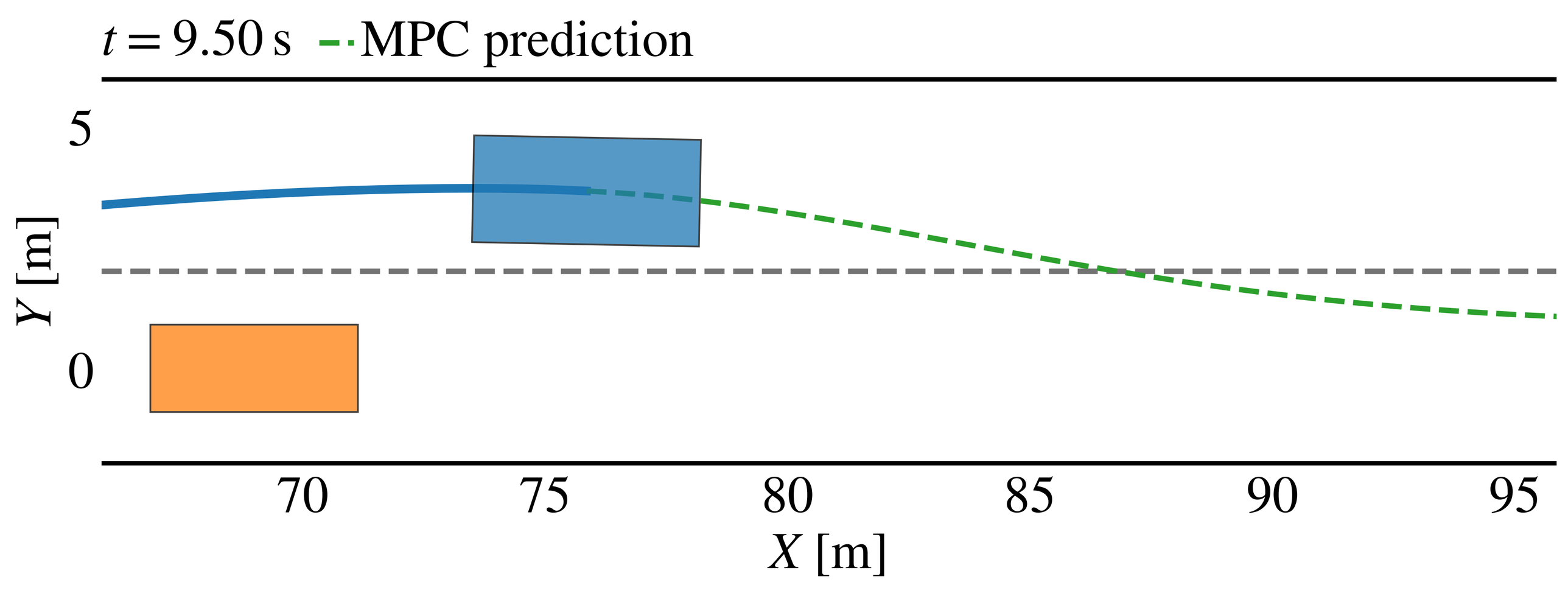

$t = 7.00$ s
MPC prediction
5
0
Y [m]
50
55
60
65
70
75
X [m]

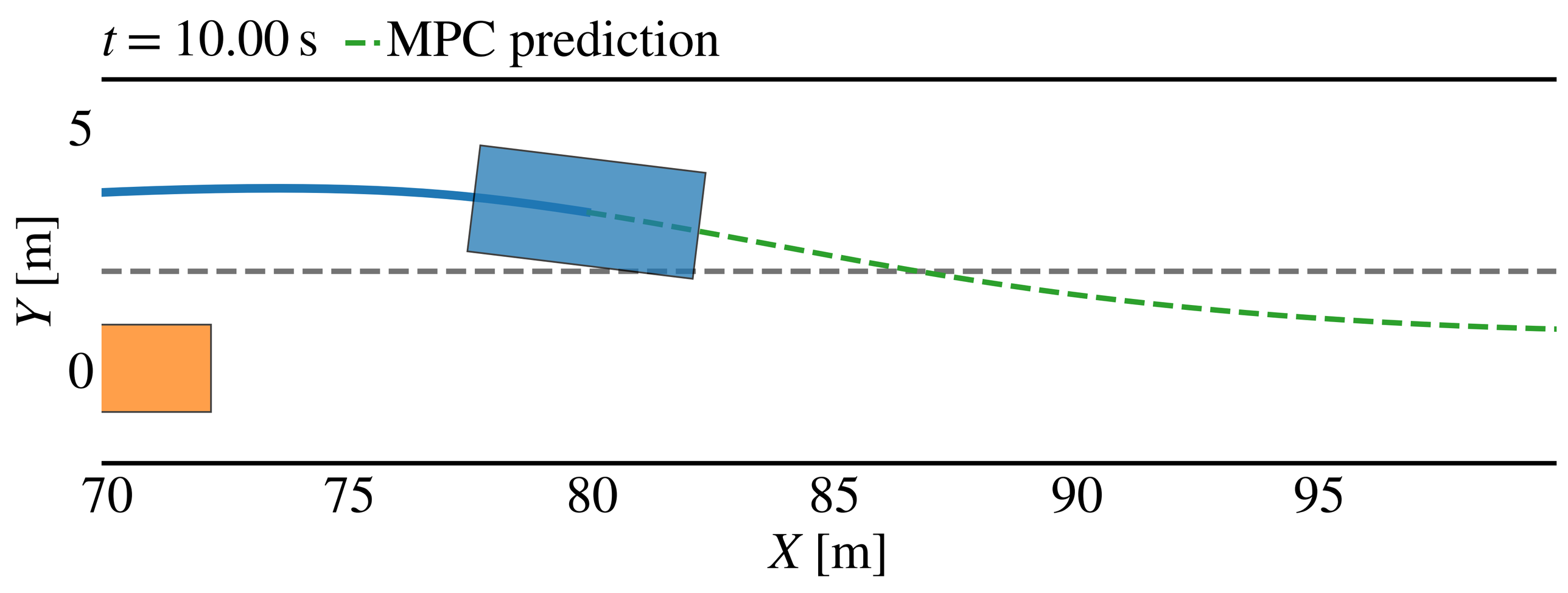

$t = 7.50$ s
MPC prediction
5
0
Y [m]
50
55
60
65
70
75
X [m]

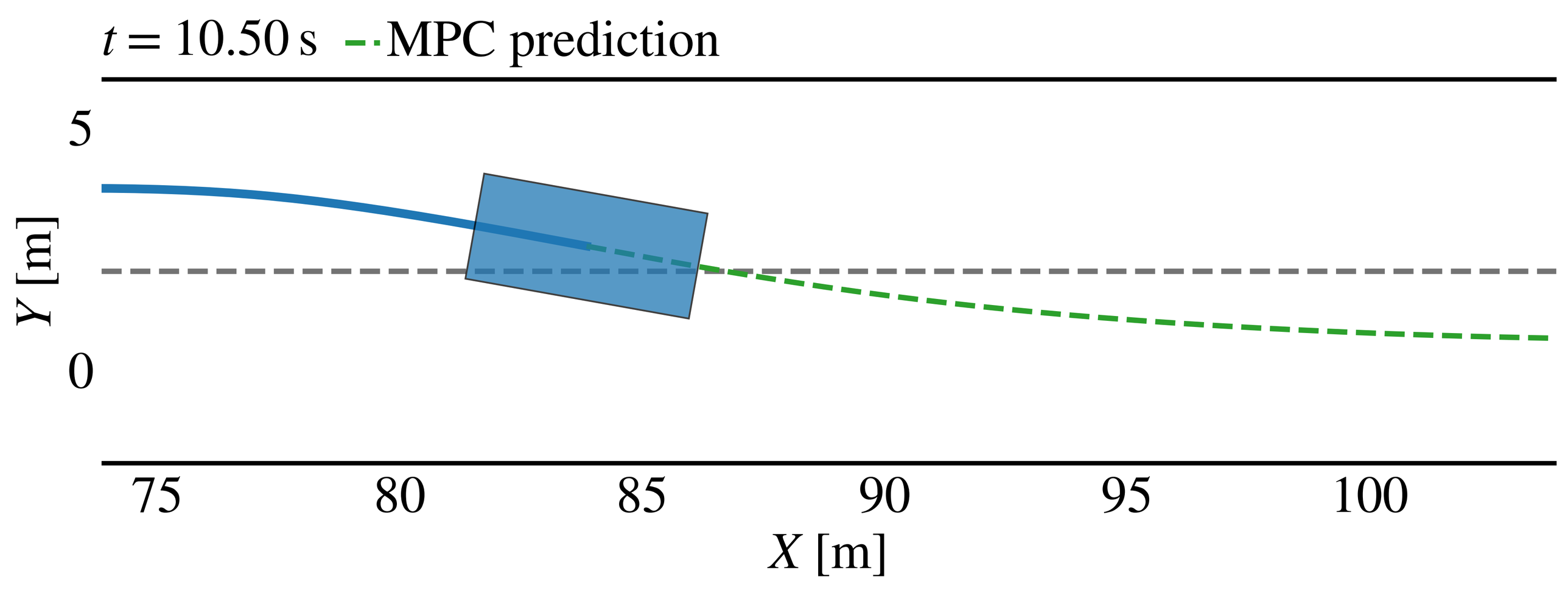

$t = 8.00$ s
MPC prediction
5
0
Y [m]
55
60
65
70
75
80
X [m]

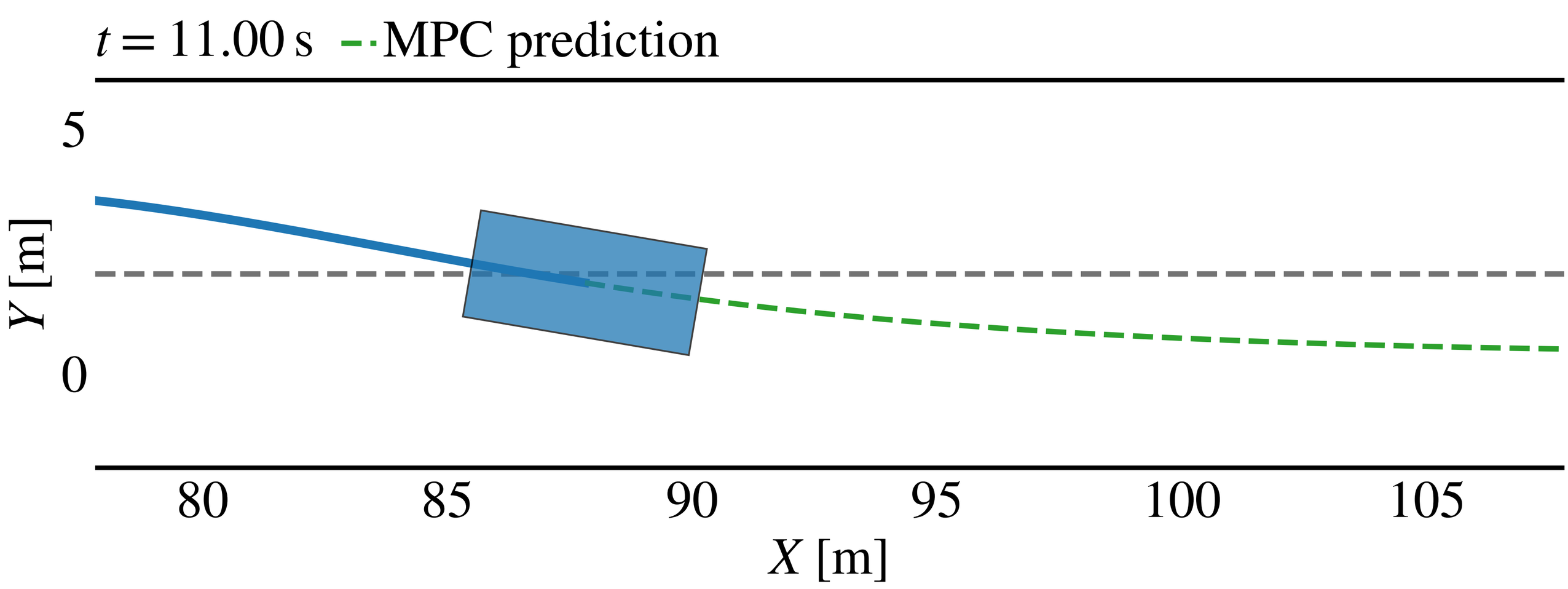

$t = 8.50$ s
MPC prediction
5
0
Y [m]
60
65
70
75
80
85
X [m]

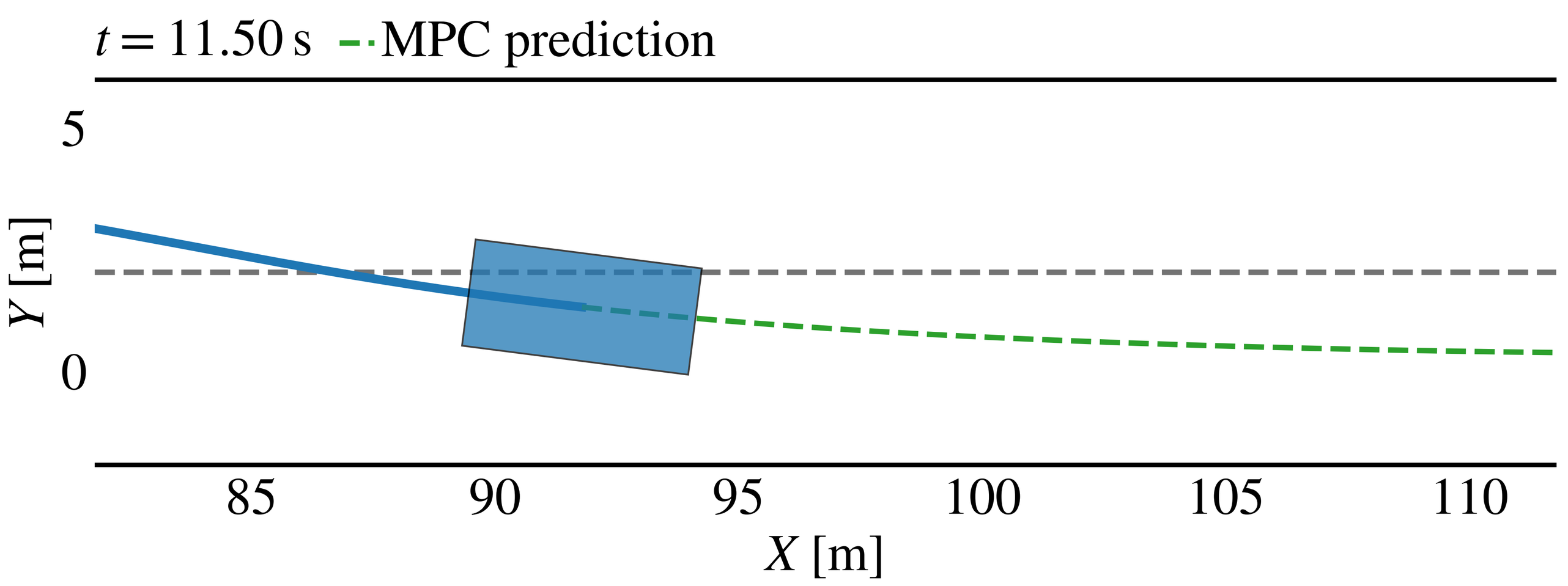

$t = 9.00$ s
MPC prediction
5
0
Y [m]
65
70
75
80
85
90
X [m]

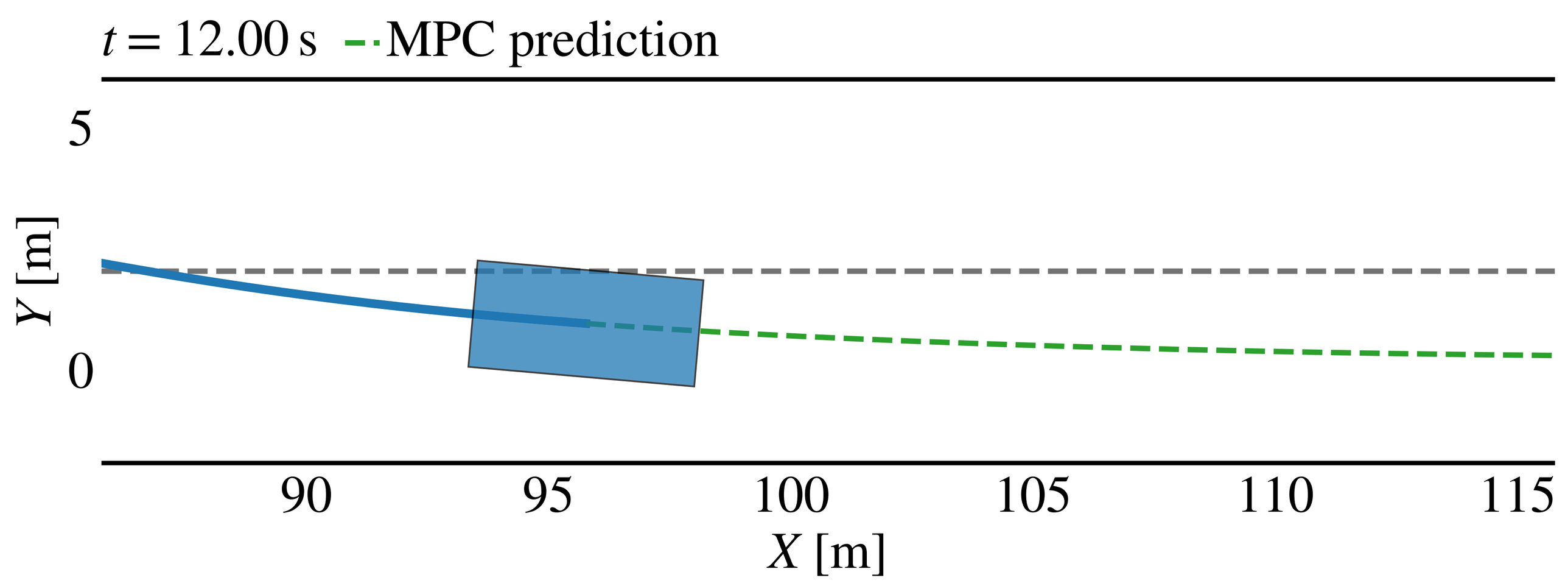

$t = 9.50$ s
MPC prediction
5
0
Y [m]
70
75
80
85
90
95
X [m]

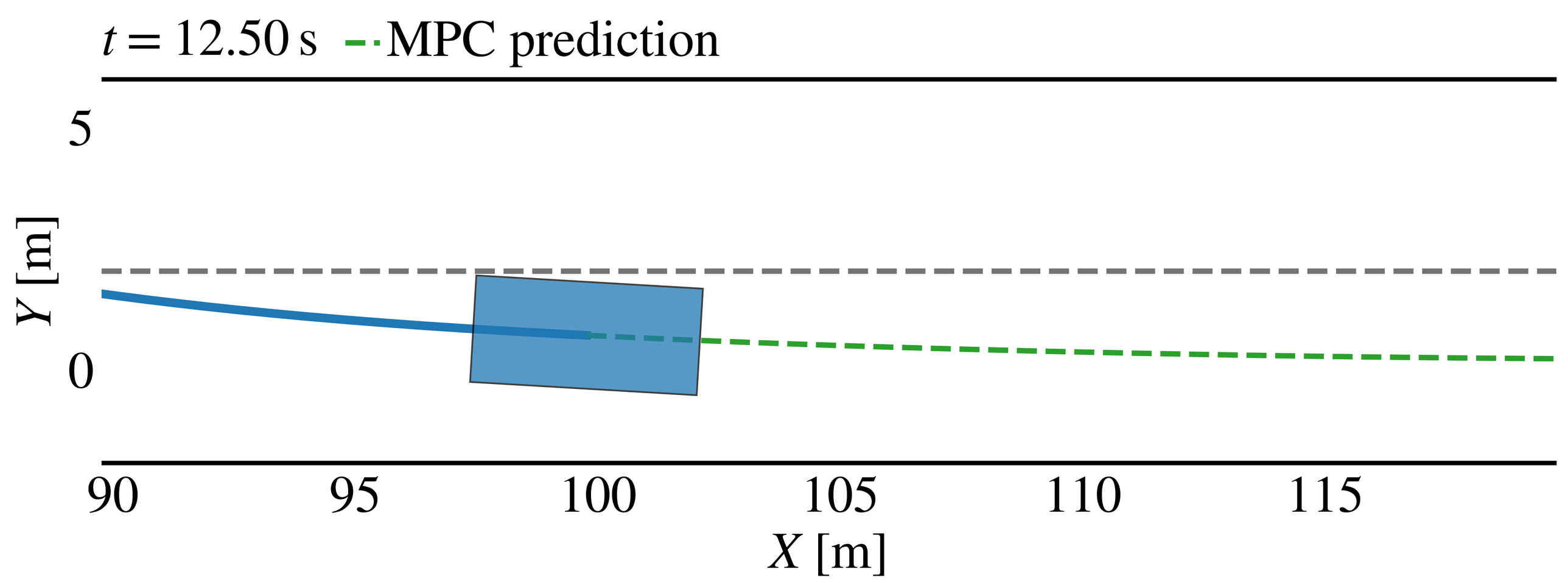

$t = 10.00$ s
MPC prediction
5
0
Y [m]
70
75
80
85
90
95
X [m]

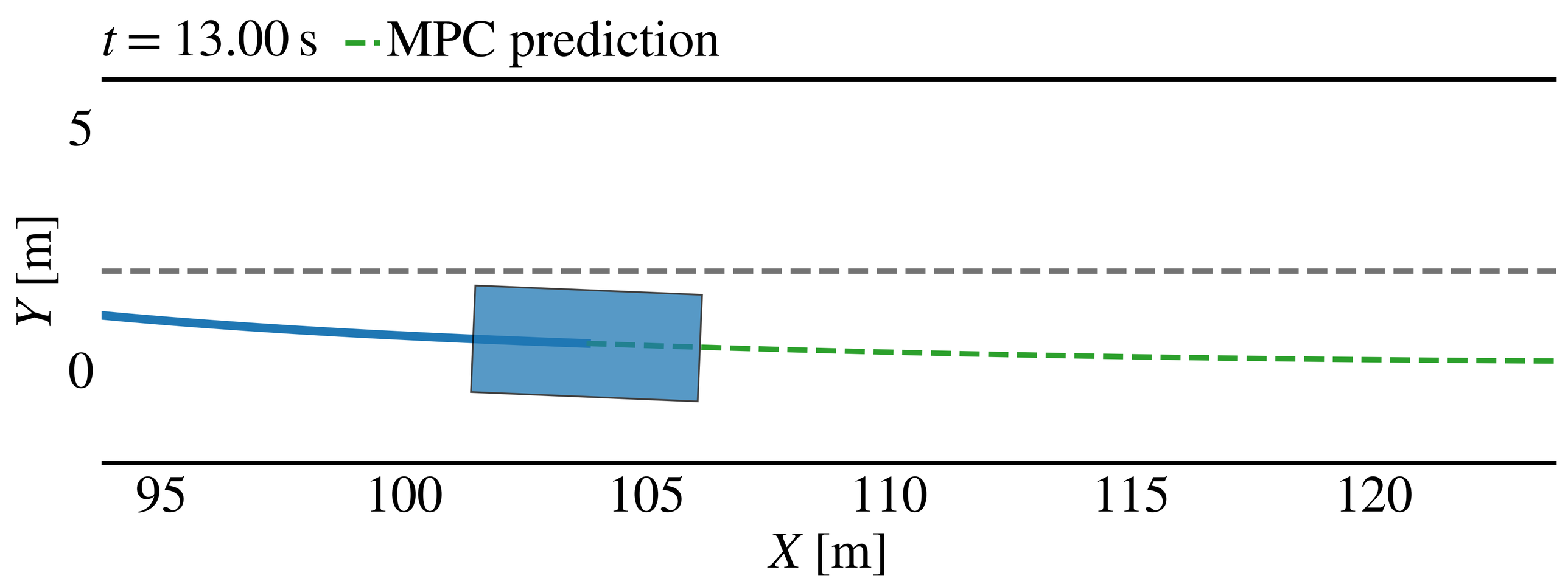

$t = 10.50$ s
MPC prediction
5
0
Y [m]
75
80
85
90
95
100
X [m]

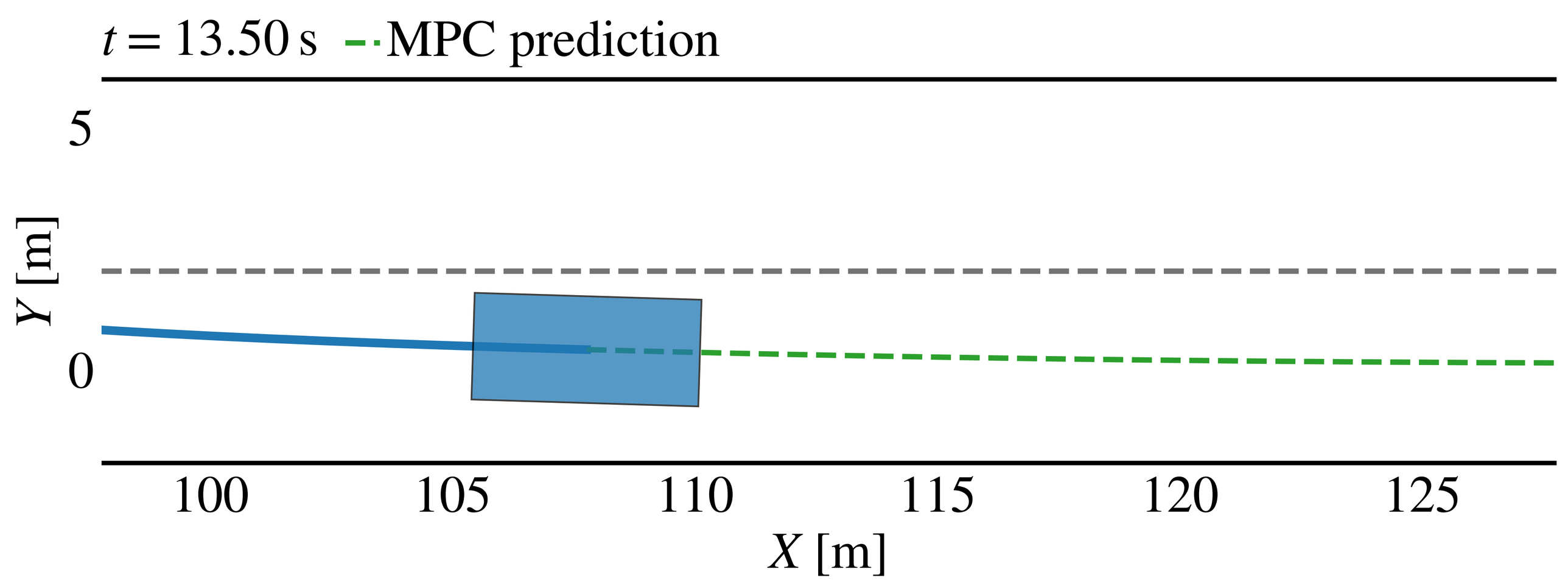

$t = 11.00$ s
MPC prediction
5
0
Y [m]
80
85
90
95
100
105
X [m]

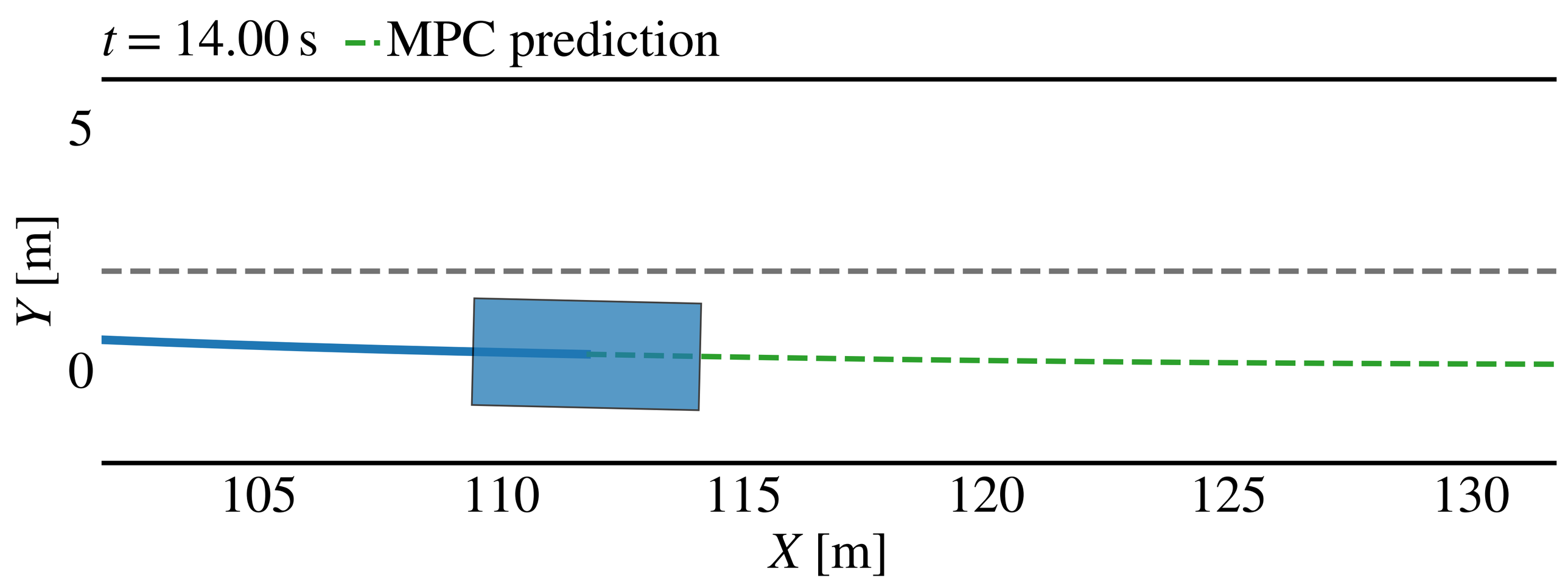

$t = 11.50$ s
MPC prediction
5
0
Y [m]
85
90
95
100
105
110
X [m]

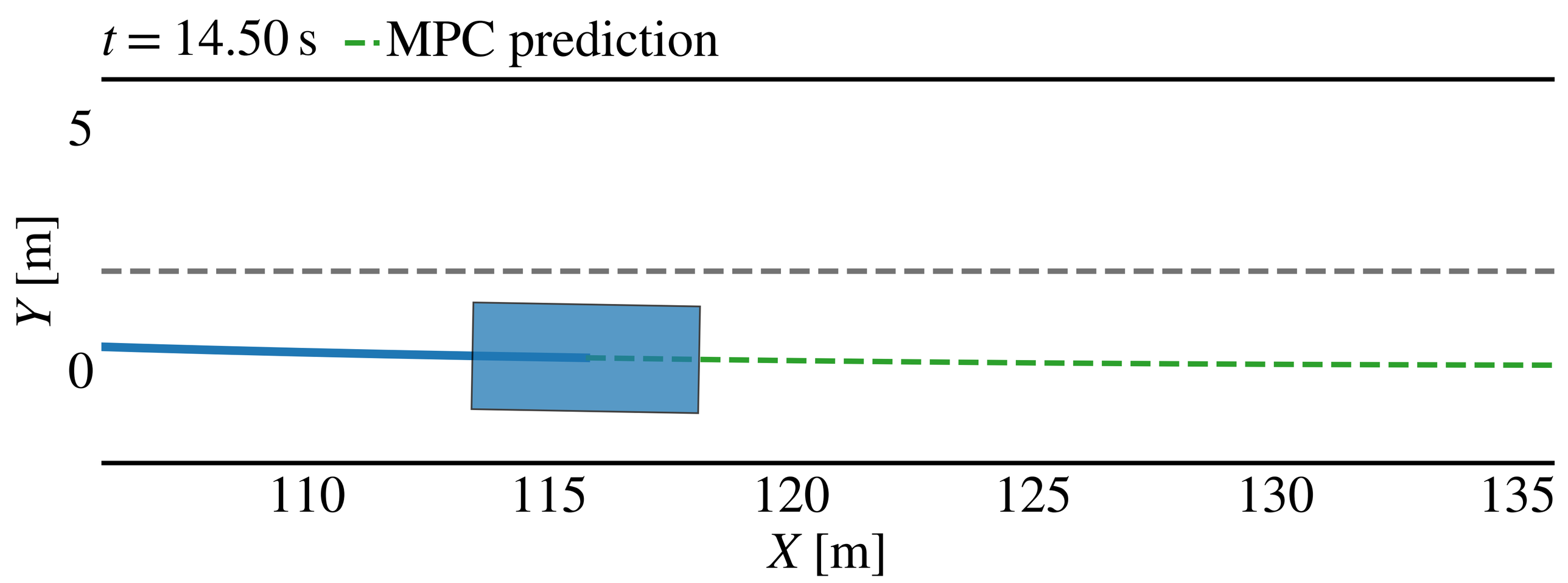

$t = 12.00$ s
MPC prediction
5
0
Y [m]
90
95
100
105
110
115
X [m]

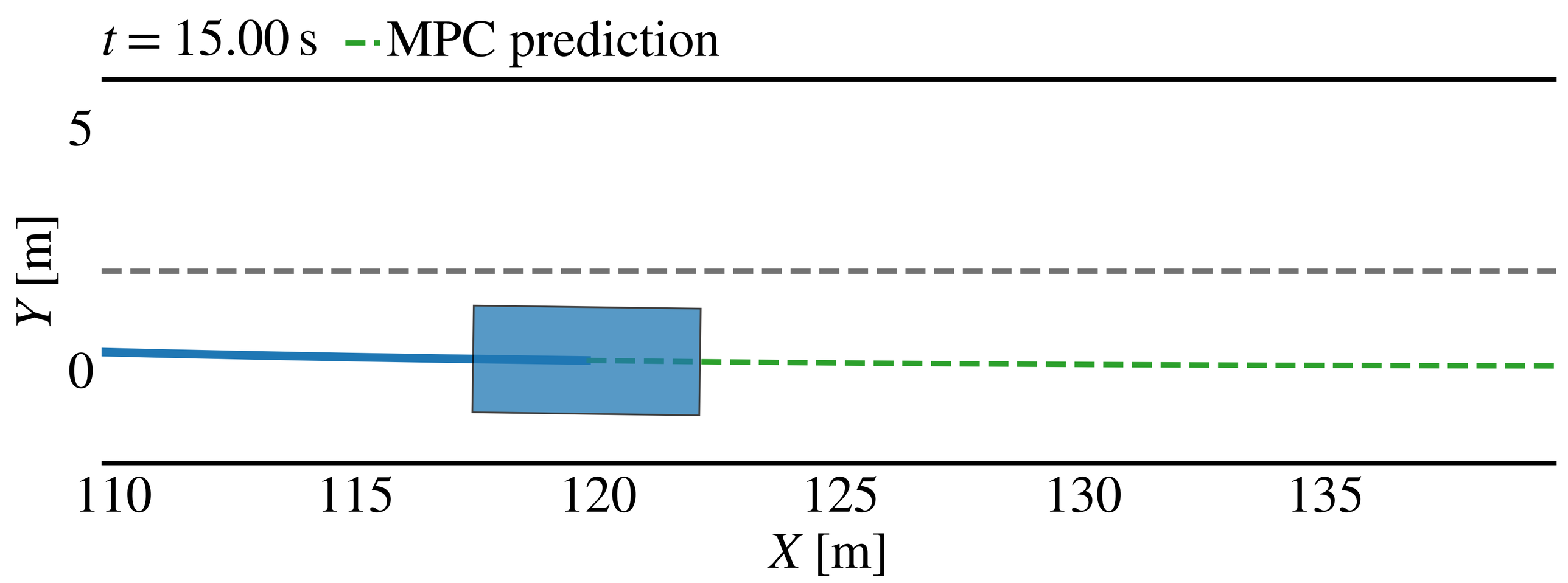

$t = 12.50$ s
MPC prediction
5
0
Y [m]
90
95
100
105
110
115
X [m]

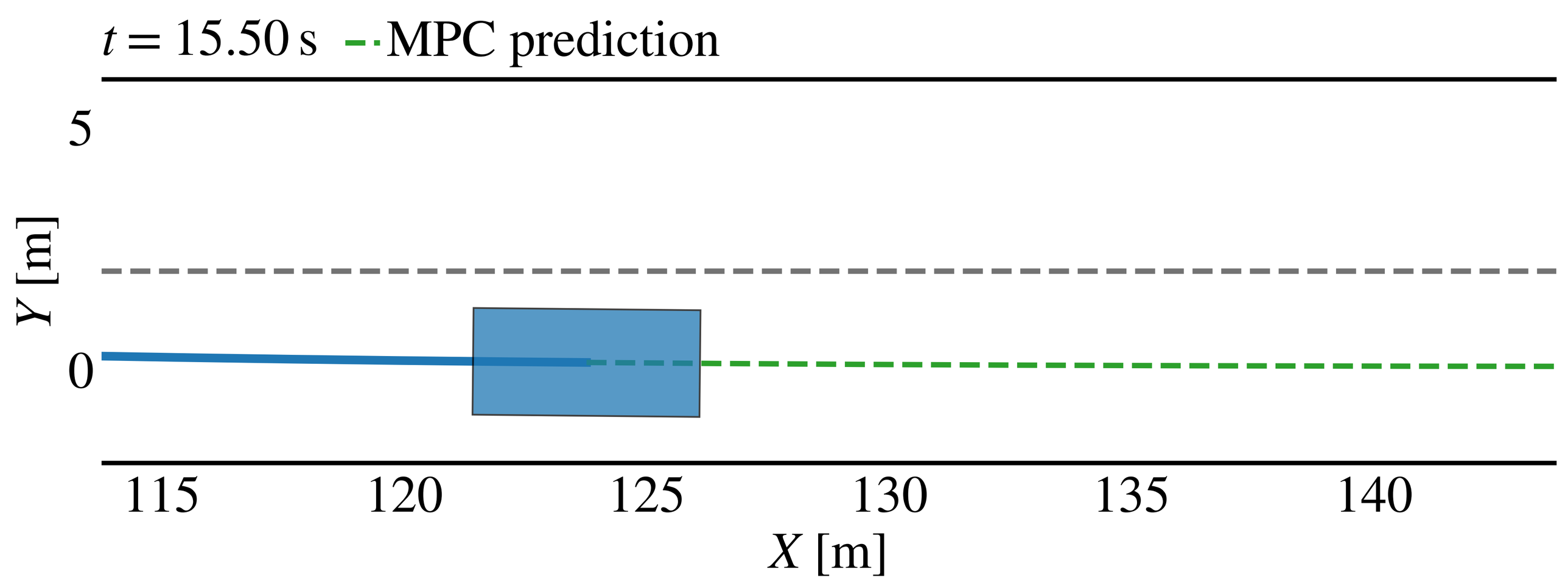

$t = 13.00$ s
MPC prediction
5
0
Y [m]
95
100
105
110
115
120
X [m]

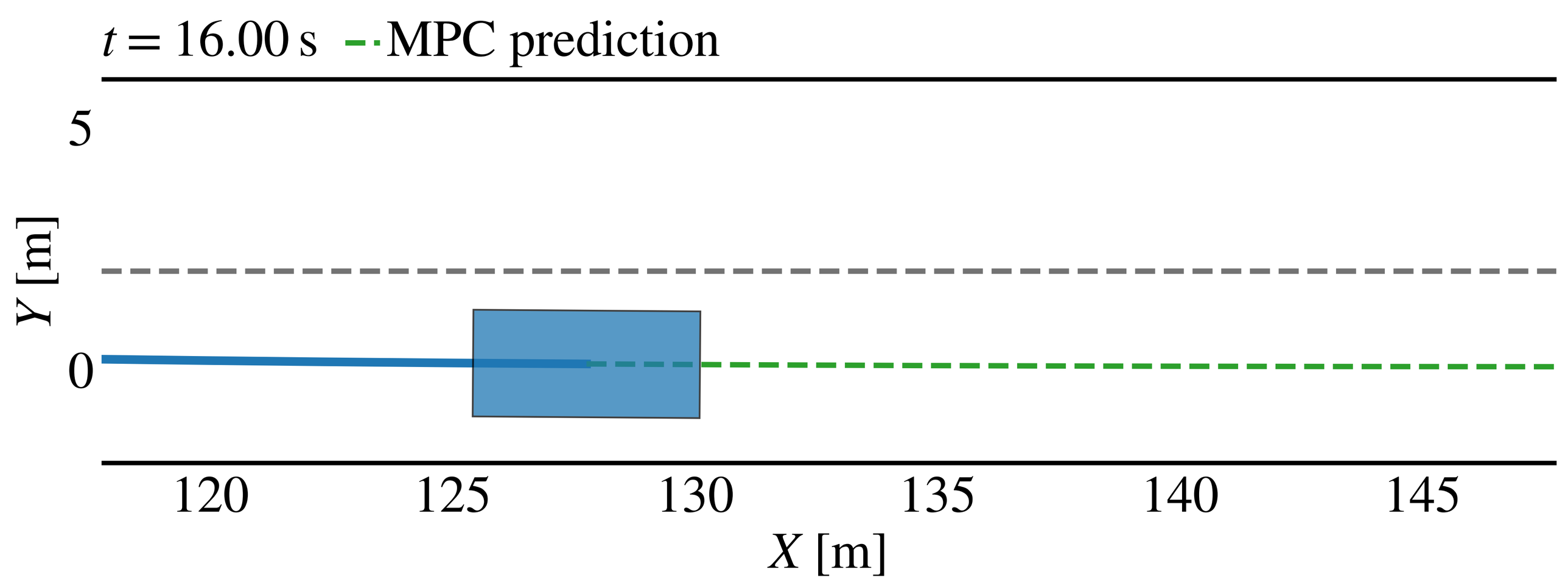

$t = 13.50$ s
MPC prediction
5
0
Y [m]
100
105
110
115
120
125
X [m]

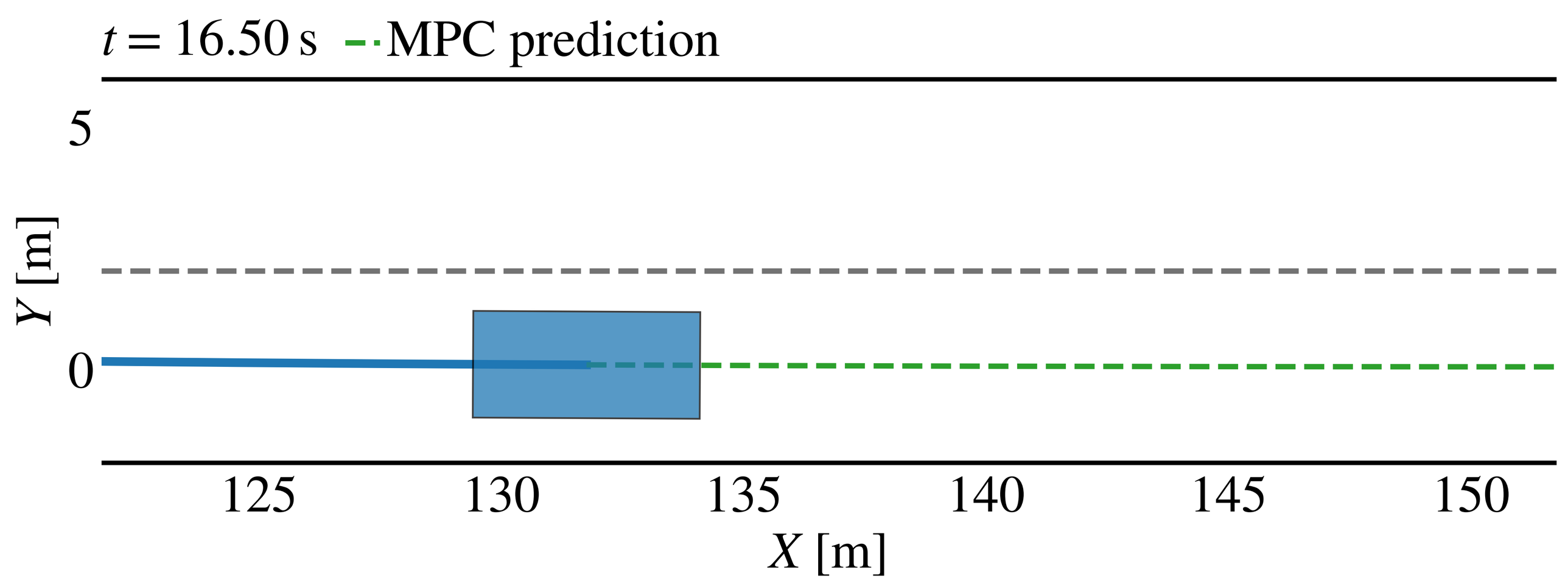

$t = 14.00$ s
MPC prediction
5
0
Y [m]
105
110
115
120
125
130
X [m]

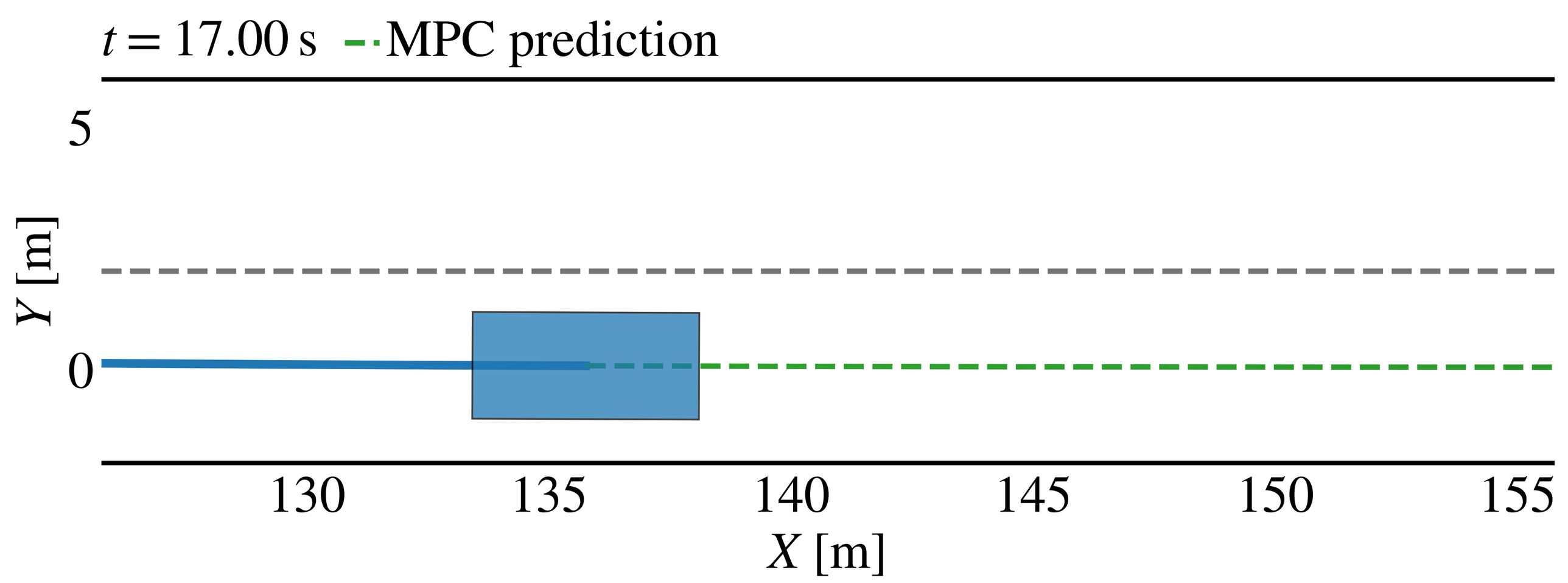

$t = 14.50$ s
MPC prediction
5
0
Y [m]
110
115
120
125
130
135
X [m]

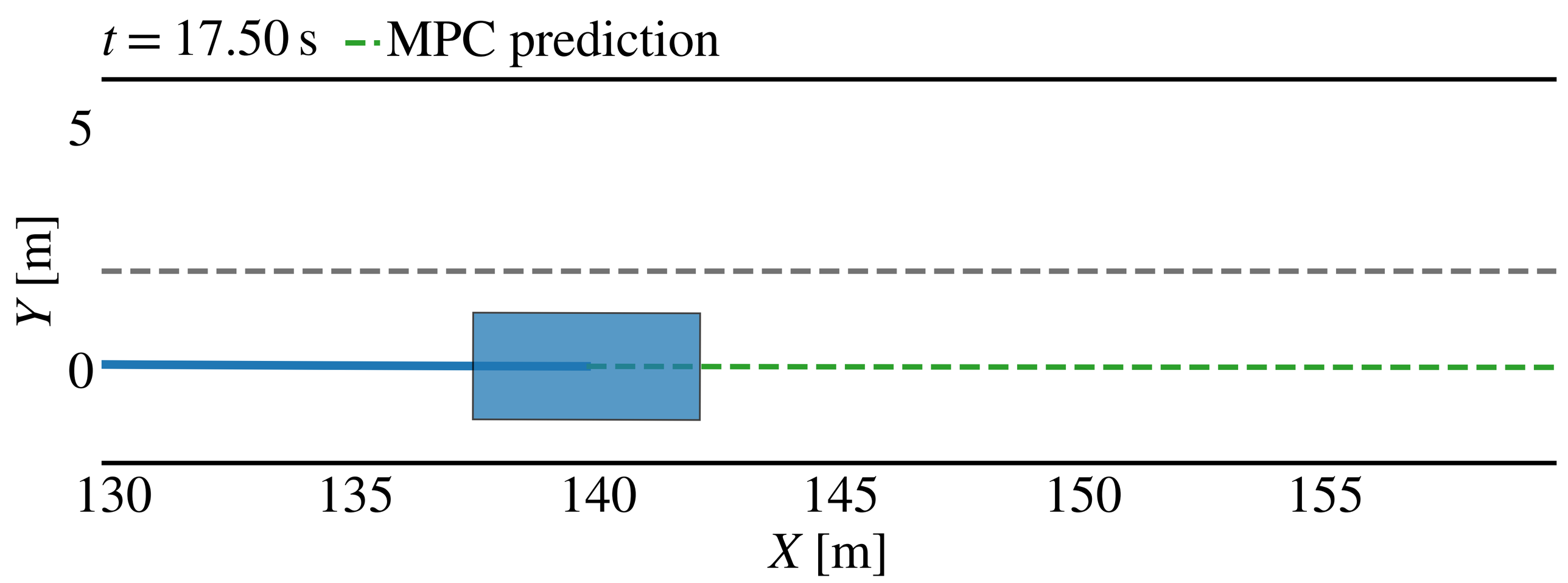

$t = 15.00$ s
MPC prediction
5
0
Y [m]
110
115
120
125
130
135
X [m]

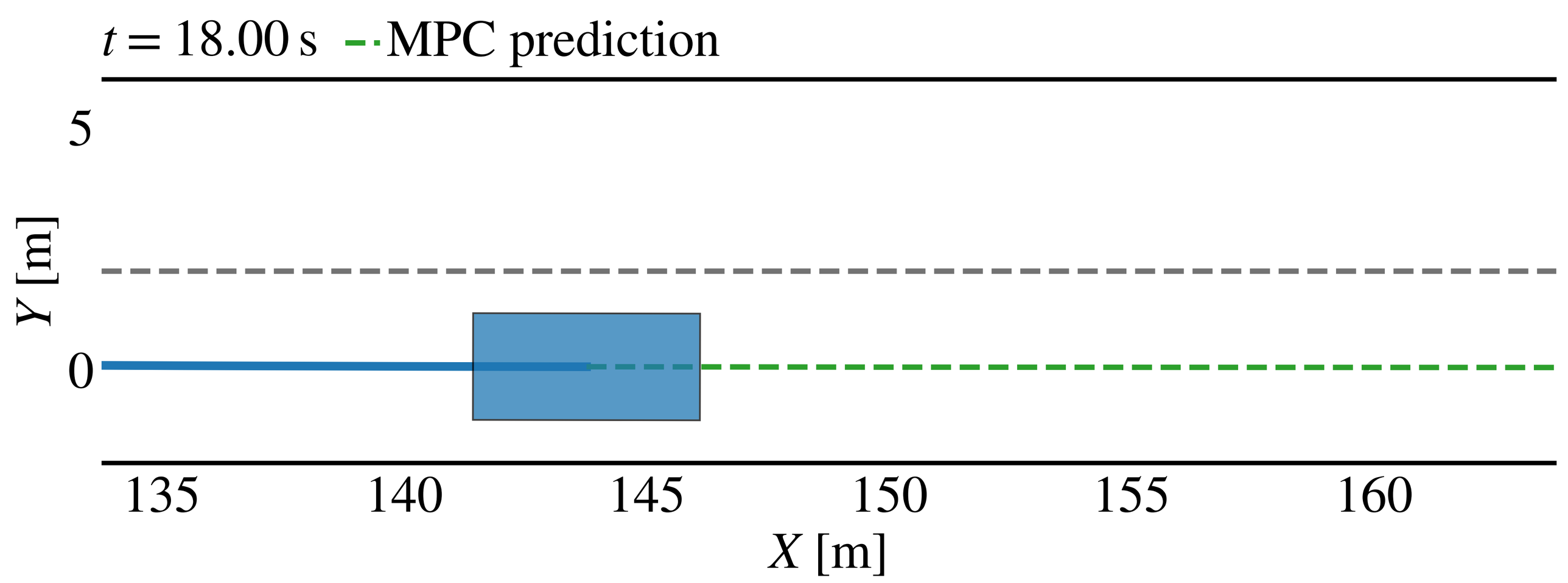

$t = 15.50$ s
MPC prediction
5
0
Y [m]
115
120
125
130
135
140
X [m]

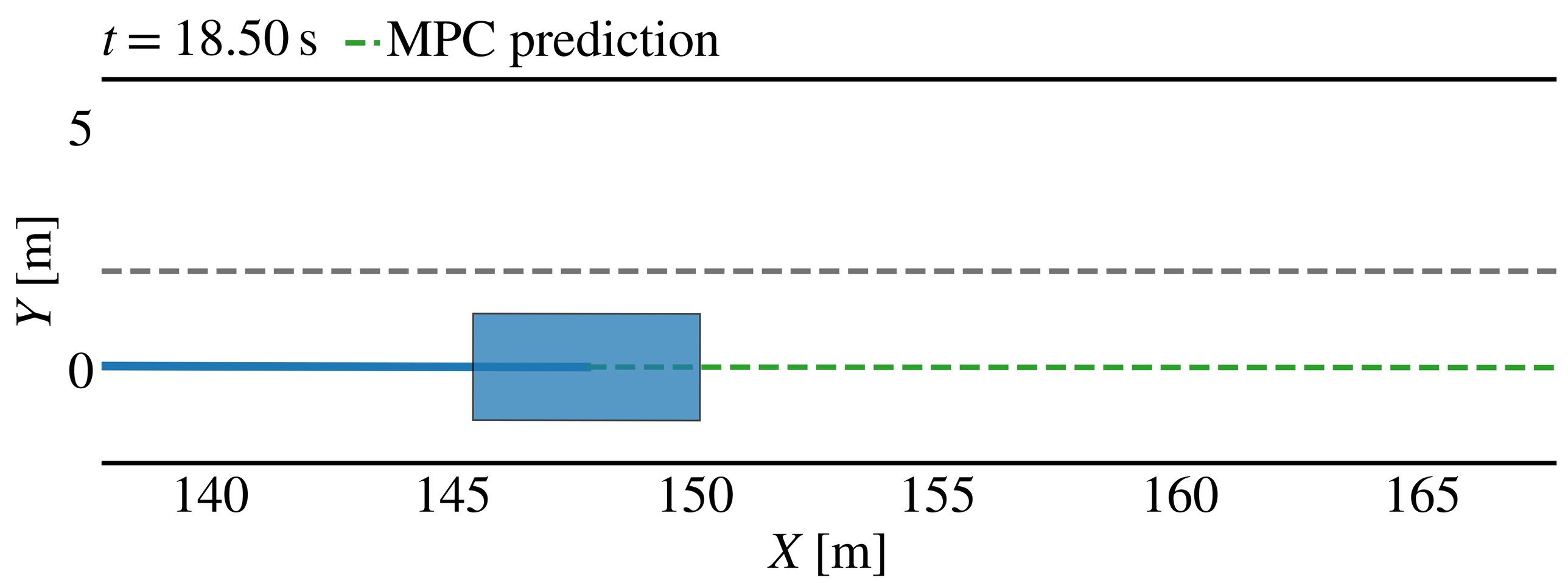

$t = 16.00$ s
MPC prediction
5
0
Y [m]
120
125
130
135
140
145
X [m]

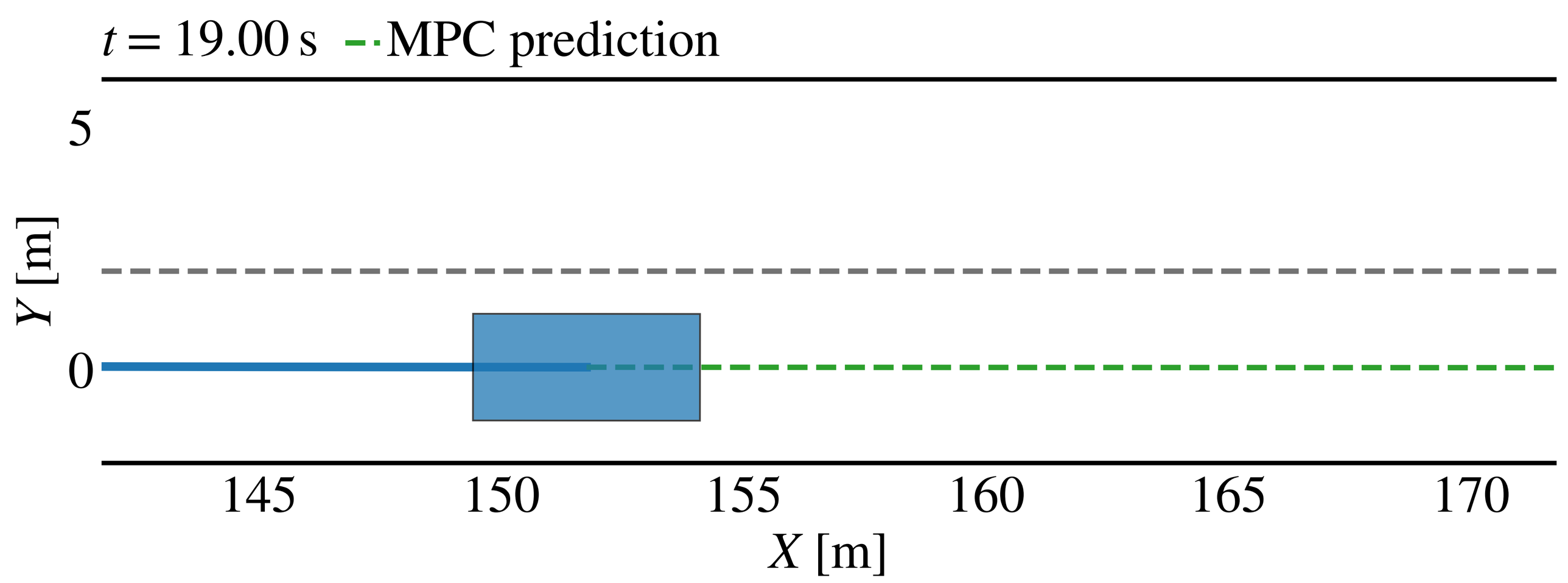

$t = 16.50$ s
MPC prediction
5
0
Y [m]
125
130
135
140
145
150
X [m]

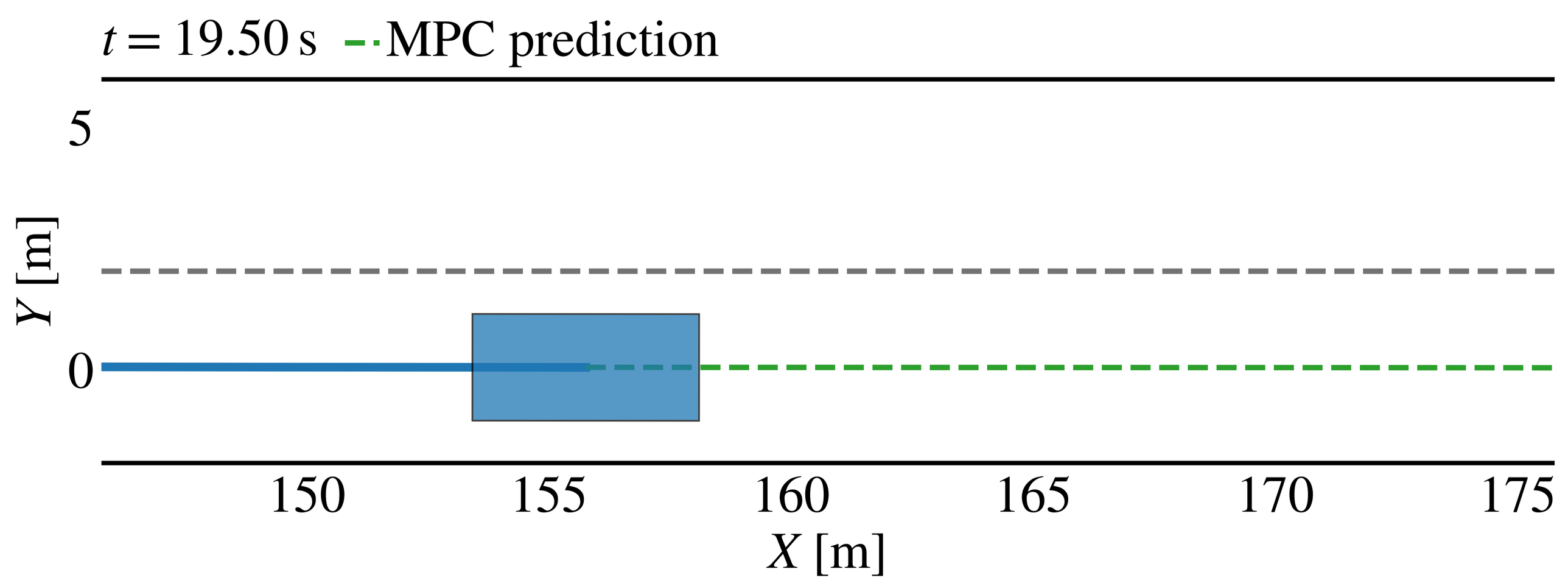

$t = 17.00$ s
MPC prediction
5
0
Y [m]
130
135
140
145
150
155
X [m]

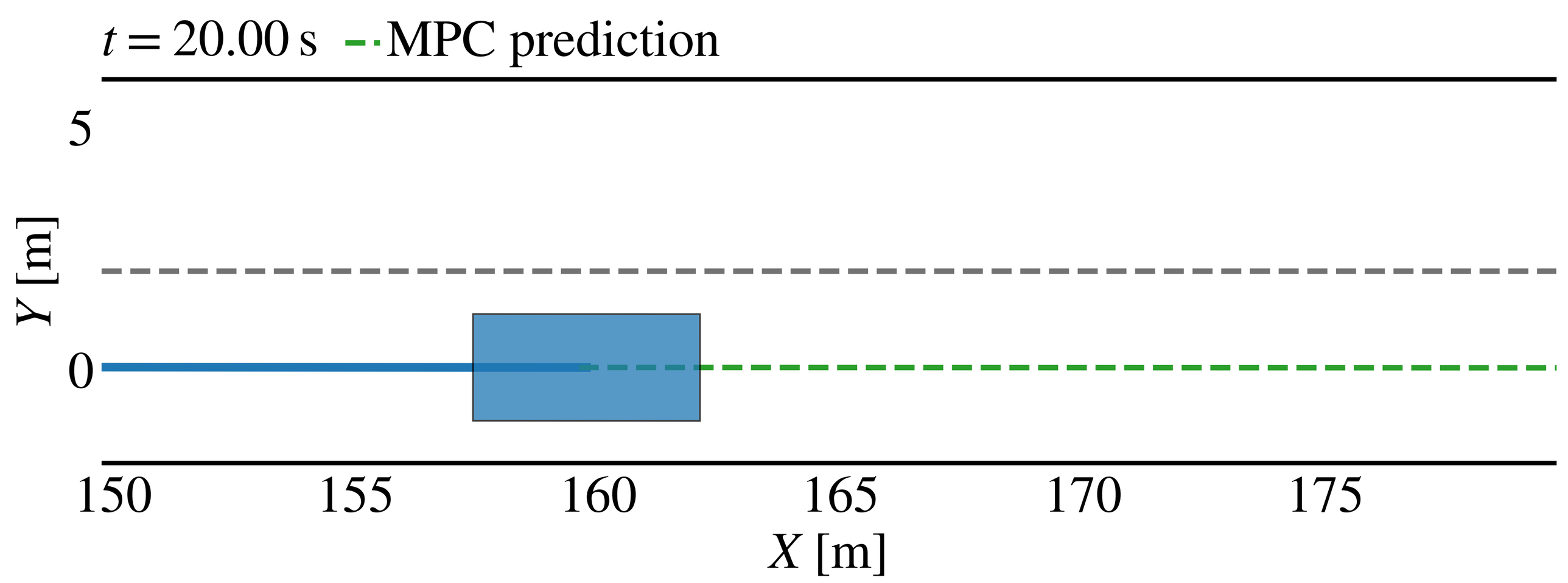

$t = 17.50$ s
MPC prediction
5
0
Y [m]
130
135
140
145
150
155
X [m]


$t = 18.00$ s
MPC prediction
5
0
Y [m]
135
140
145
150
155
160
X [m]


$t = 18.50$ s
MPC prediction
5
0
Y [m]
140
145
150
155
160
165
X [m]


$t = 19.00$ s
MPC prediction
5
0
Y [m]
145
150
155
160
165
170
X [m]


$t = 19.50$ s
MPC prediction
5
0
Y [m]
150
155
160
165
170
175
X [m]


$t = 20.00$ s
MPC prediction
5
0
Y [m]
150
155
160
165
170
175
X [m]